



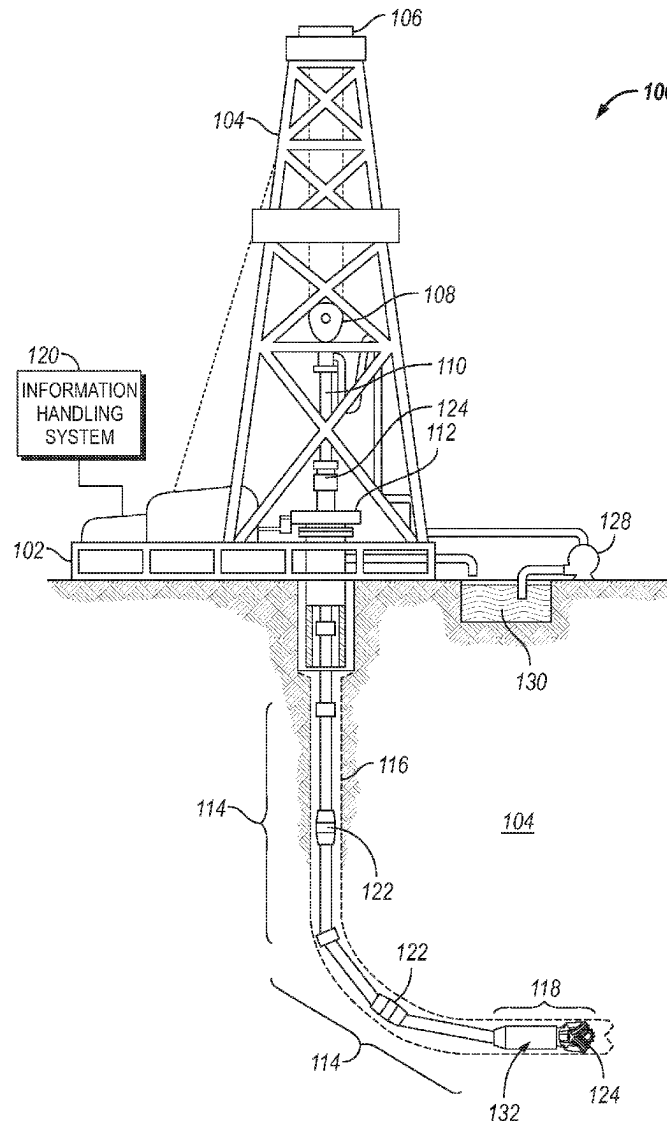
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(19) **United States**(12) **Patent Application Publication**
Wang et al.(10) **Pub. No.: US 2025/0284024 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **GAMMA-RAY SPECTROSCOPY LOGGING TOOL****Publication Classification**(51) **Int. Cl.****G01V 5/10** (2006.01)**G01V 5/04** (2006.01)(52) **U.S. Cl.****CPC** **G01V 5/102** (2013.01); **G01V 5/045** (2013.01)(71) Applicant: **Halliburton Energy Services, Inc.**,
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Houston, TX (US)(21) Appl. No.: **18/598,212**(22) Filed: **Mar. 7, 2024**

(57)

ABSTRACT

A variety of methods and apparatus are disclosed, including, in one embodiment, a pulsed neutron logging tool for a borehole in a subterranean formation, including: a pulsed neutron generator to broadcast neutrons into the subterranean formation; and a gamma ray scintillator detector comprising cerium bromide (CeBr_3), wherein the pulsed neutron logging tool (e.g., as lowered into the borehole) is configured to detect neutron-induced gamma rays from the subterranean formation and natural gamma rays from the subterranean formation.



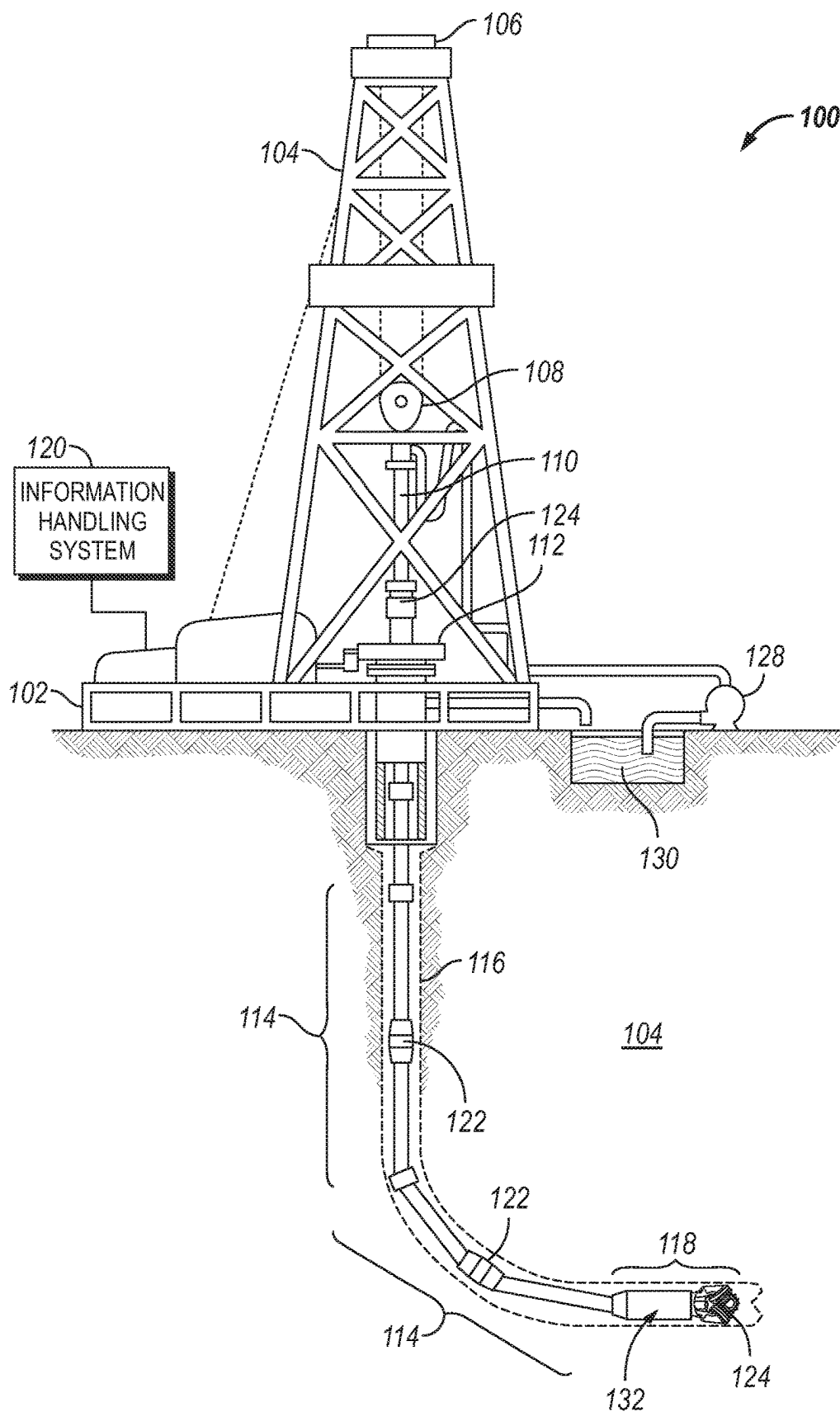


Figure 1

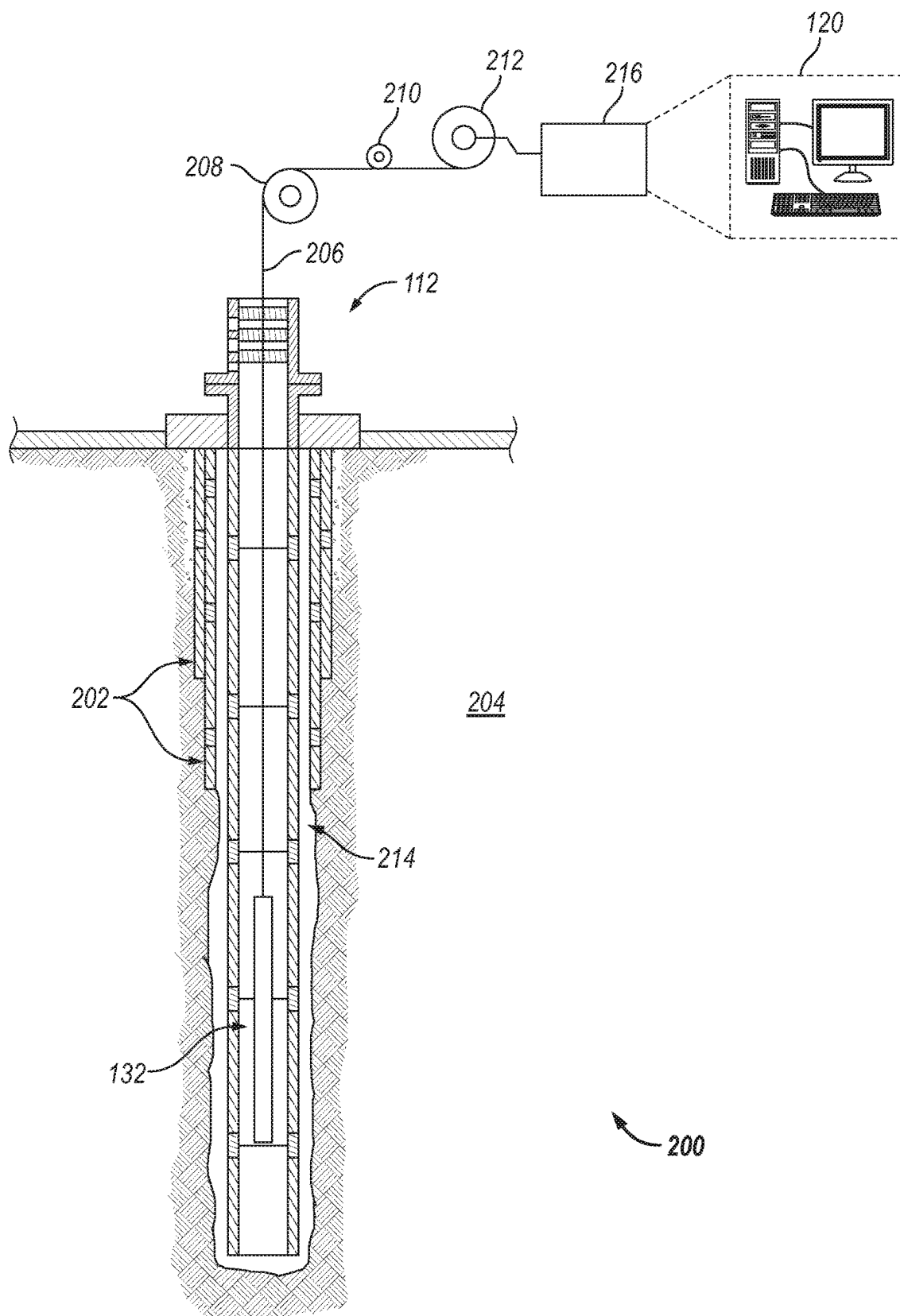


Figure 2

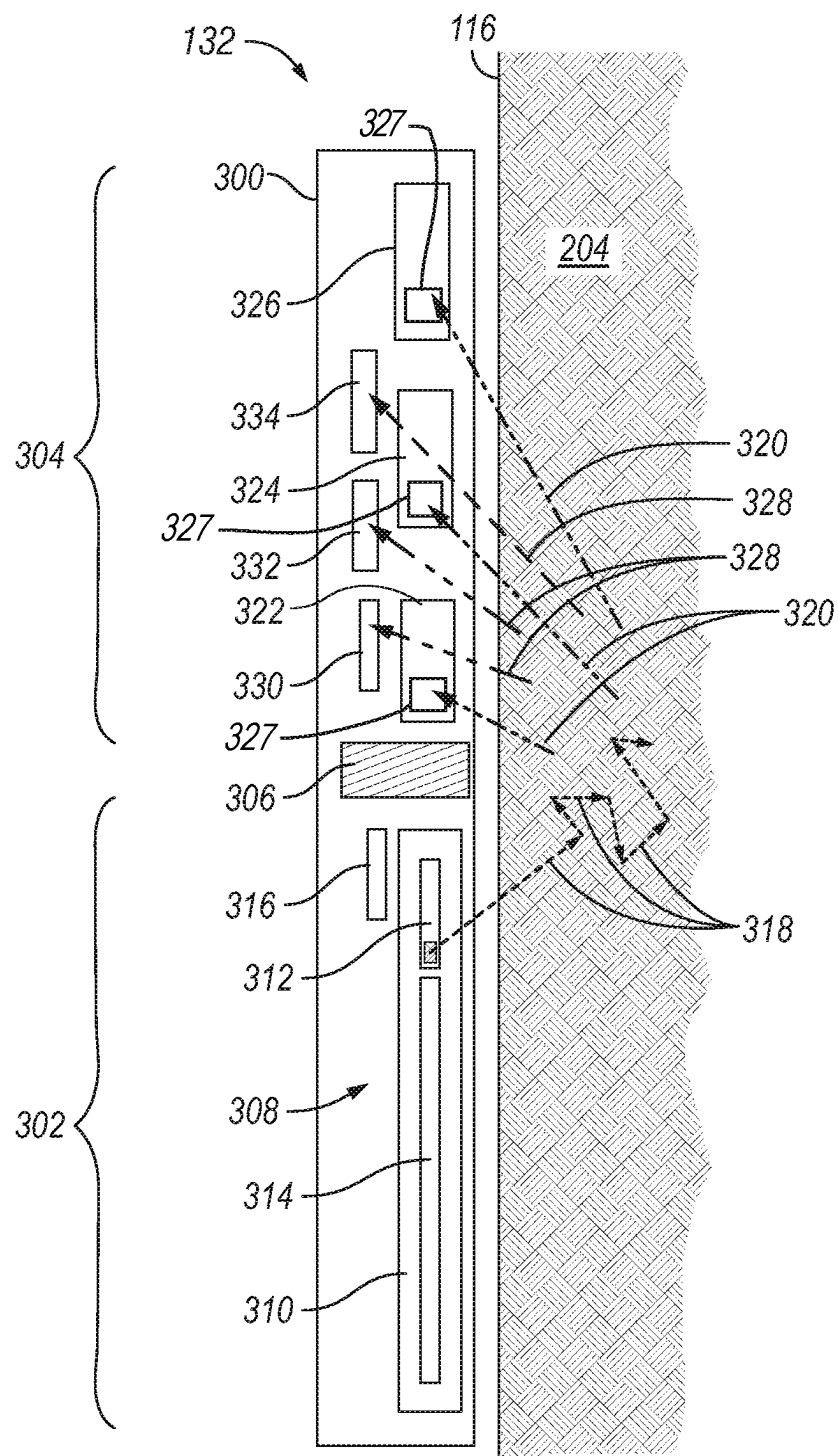


Figure 3

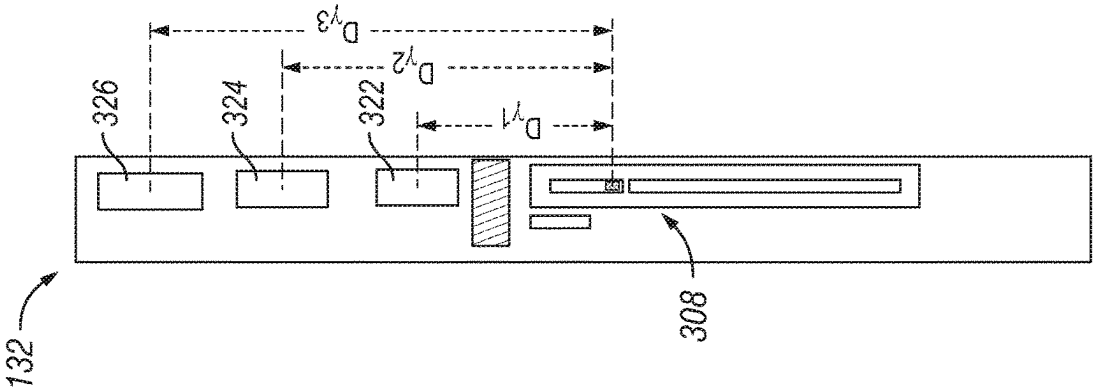


Figure 4C

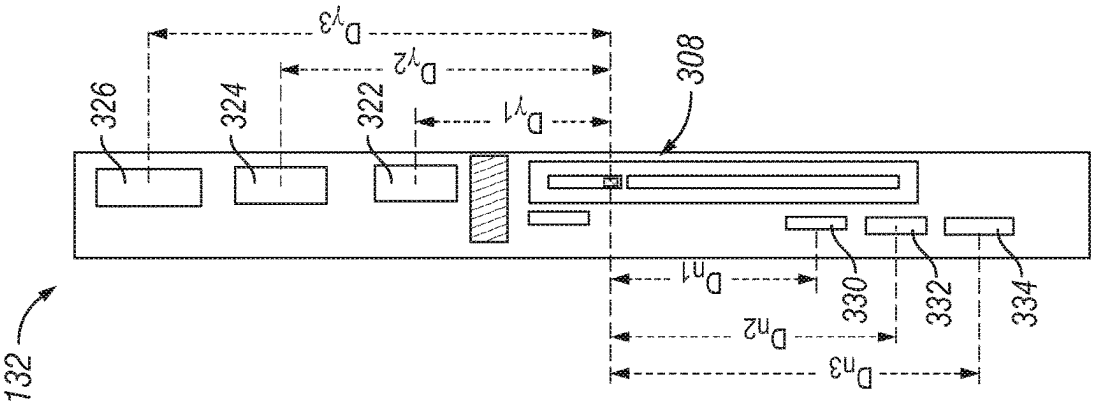


Figure 4B

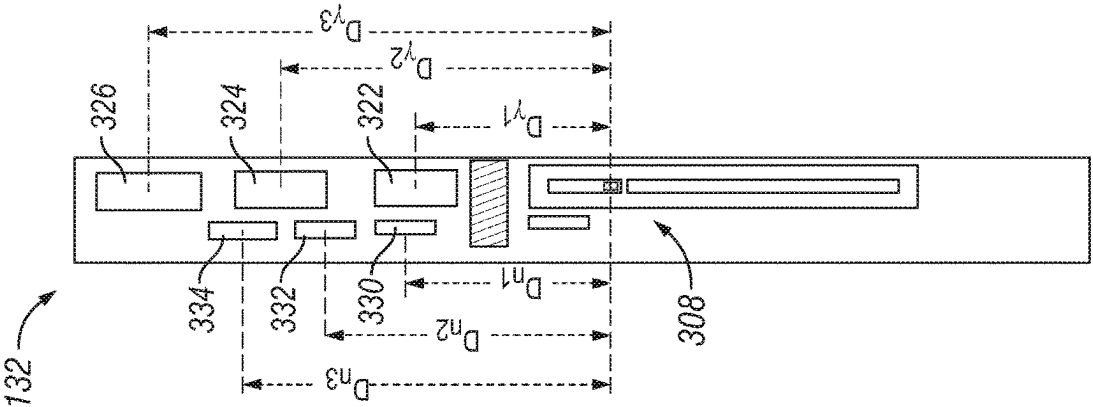


Figure 4A

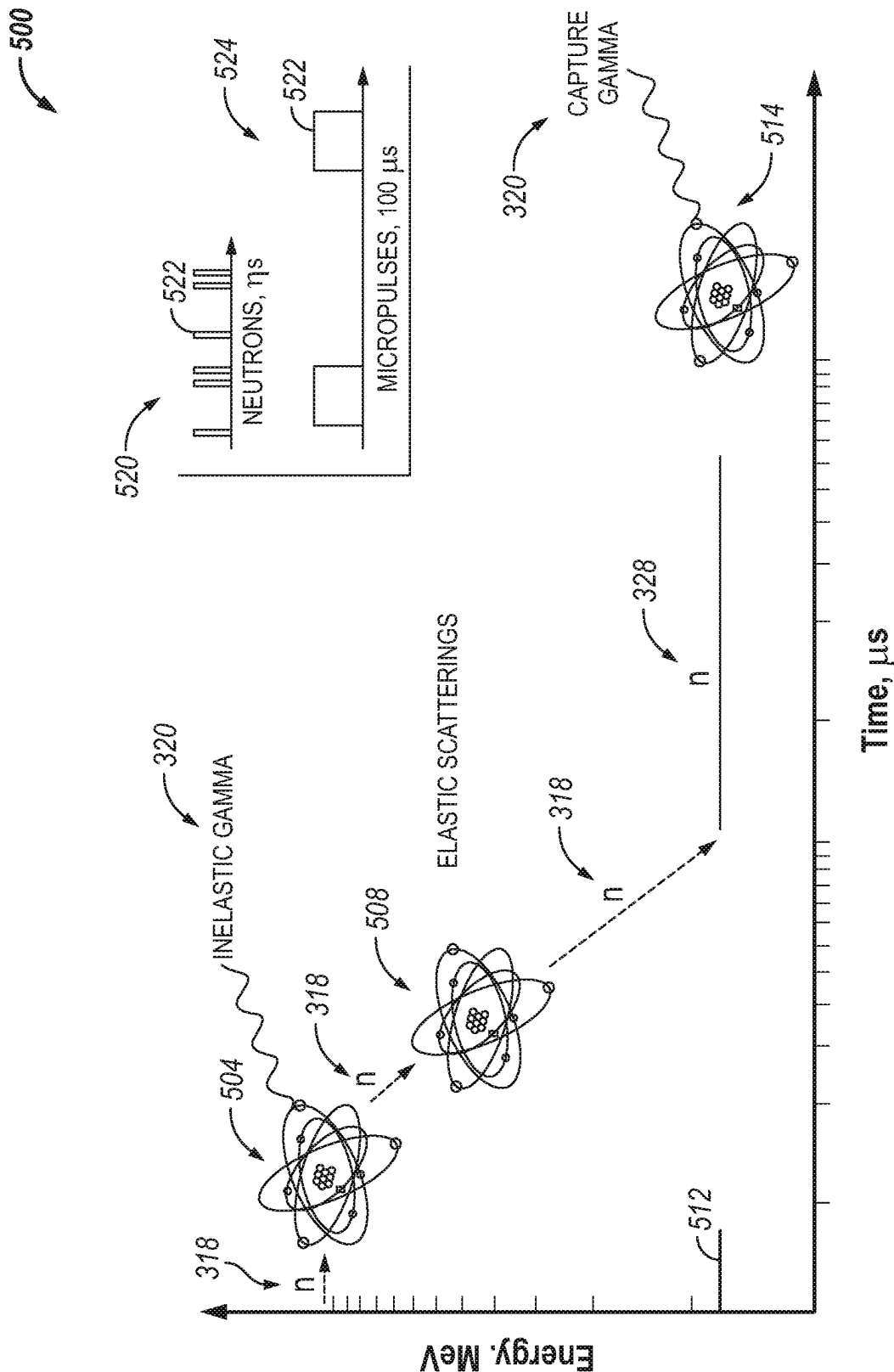


Figure 5

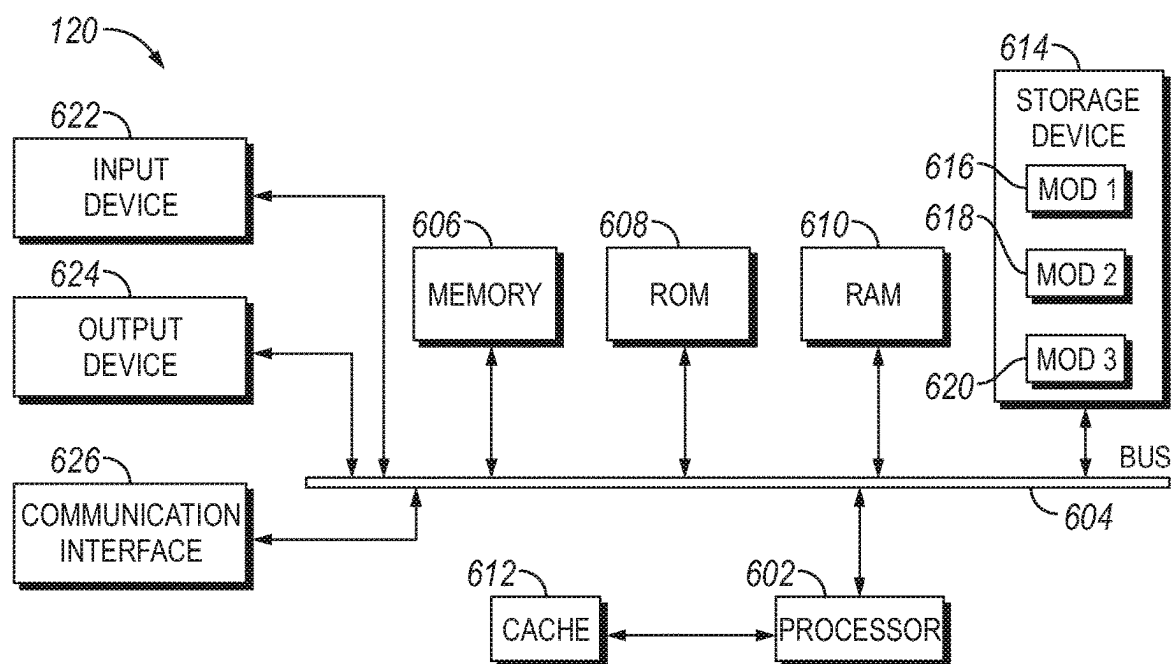


Figure 6

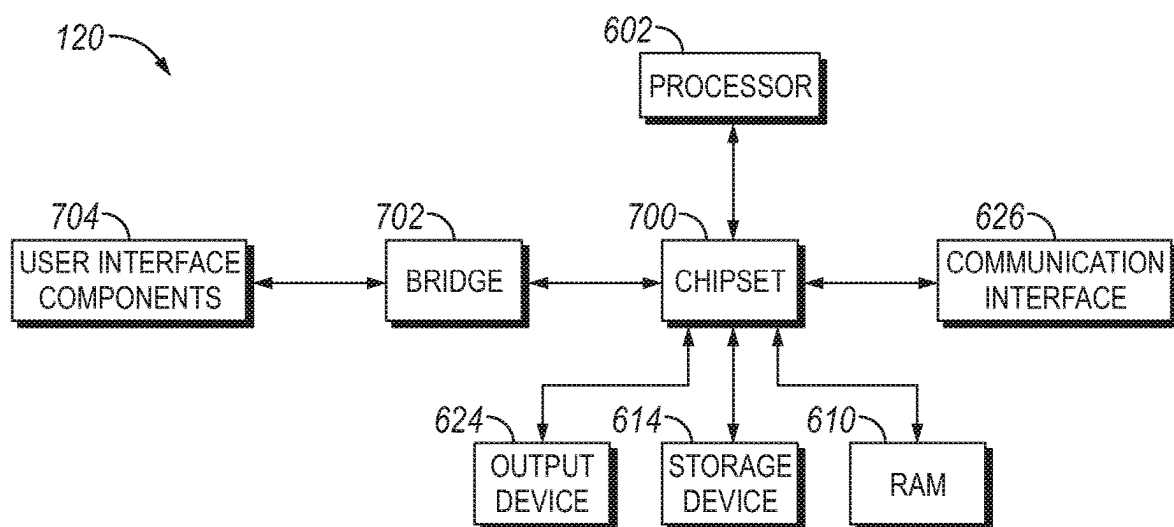


Figure 7

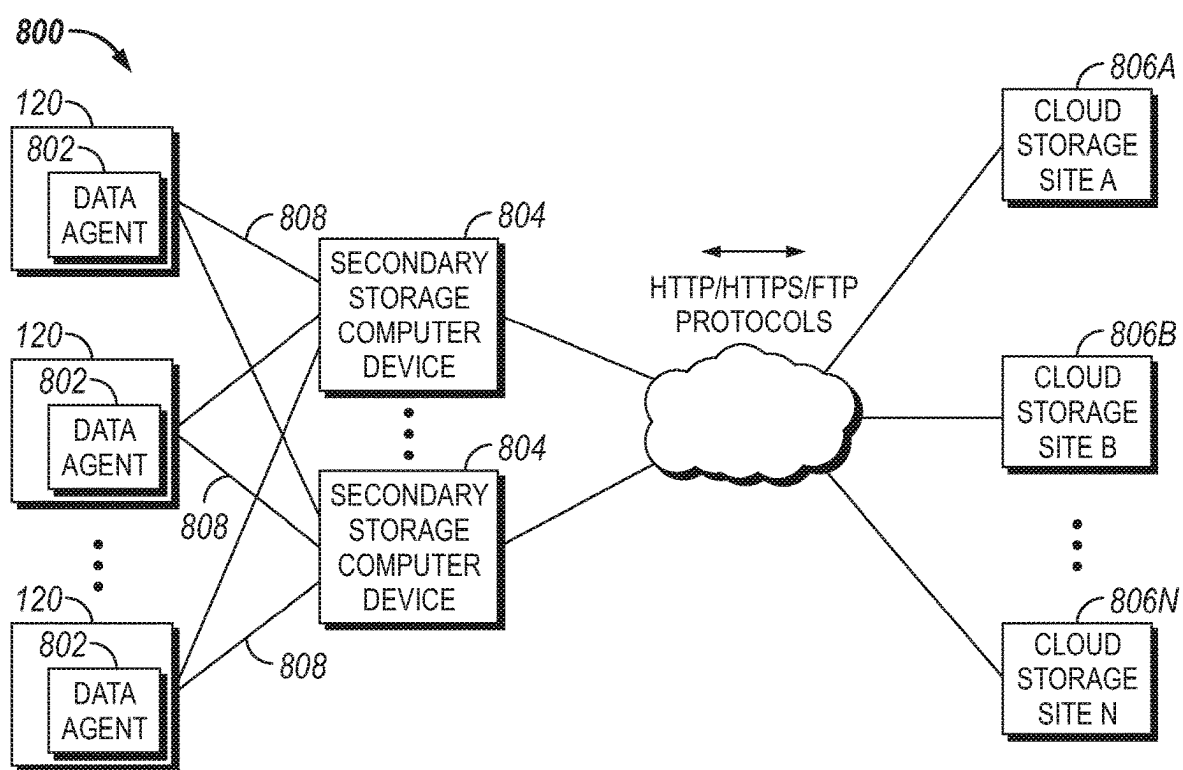


Figure 8

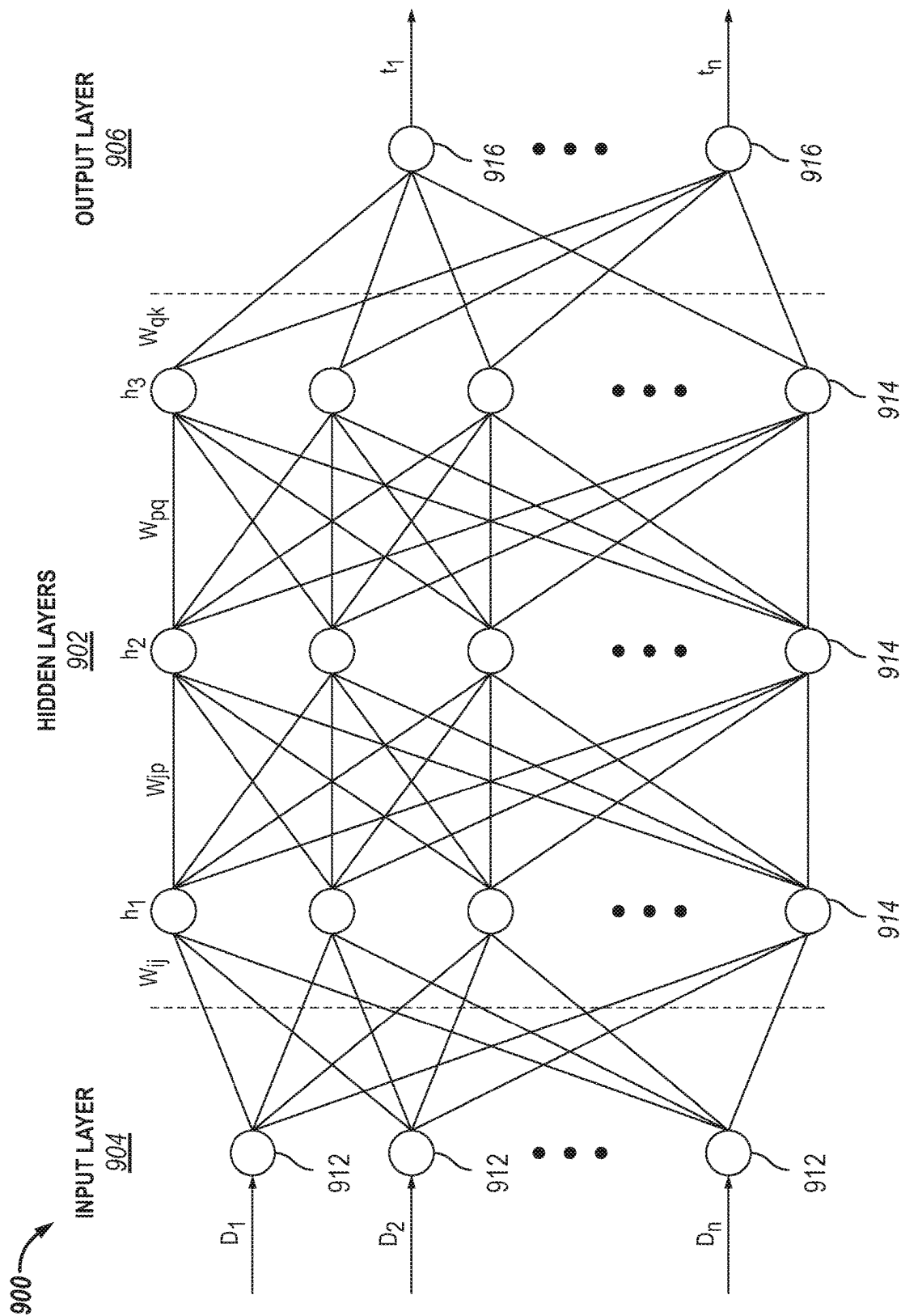


Figure 9

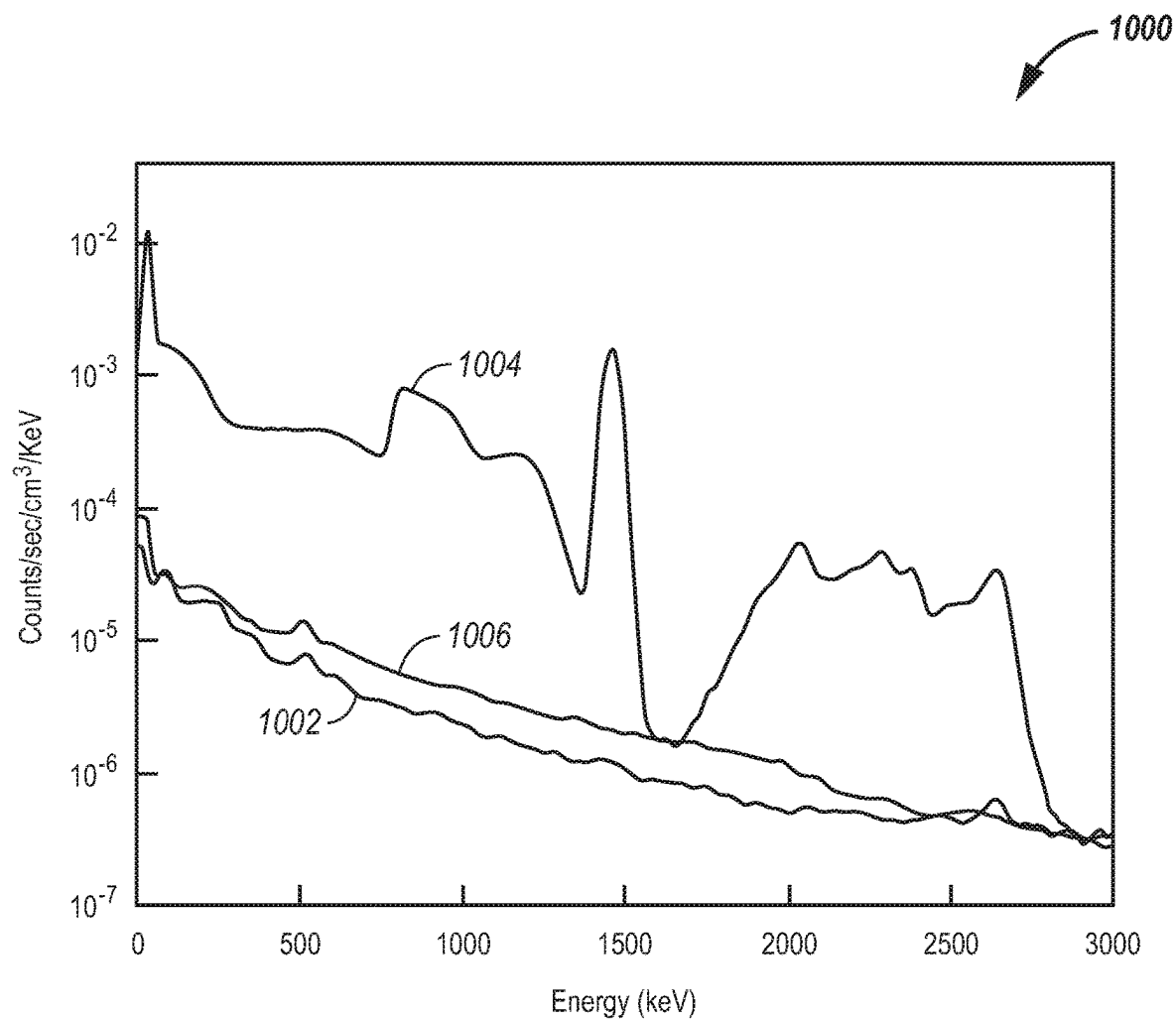


Figure 10

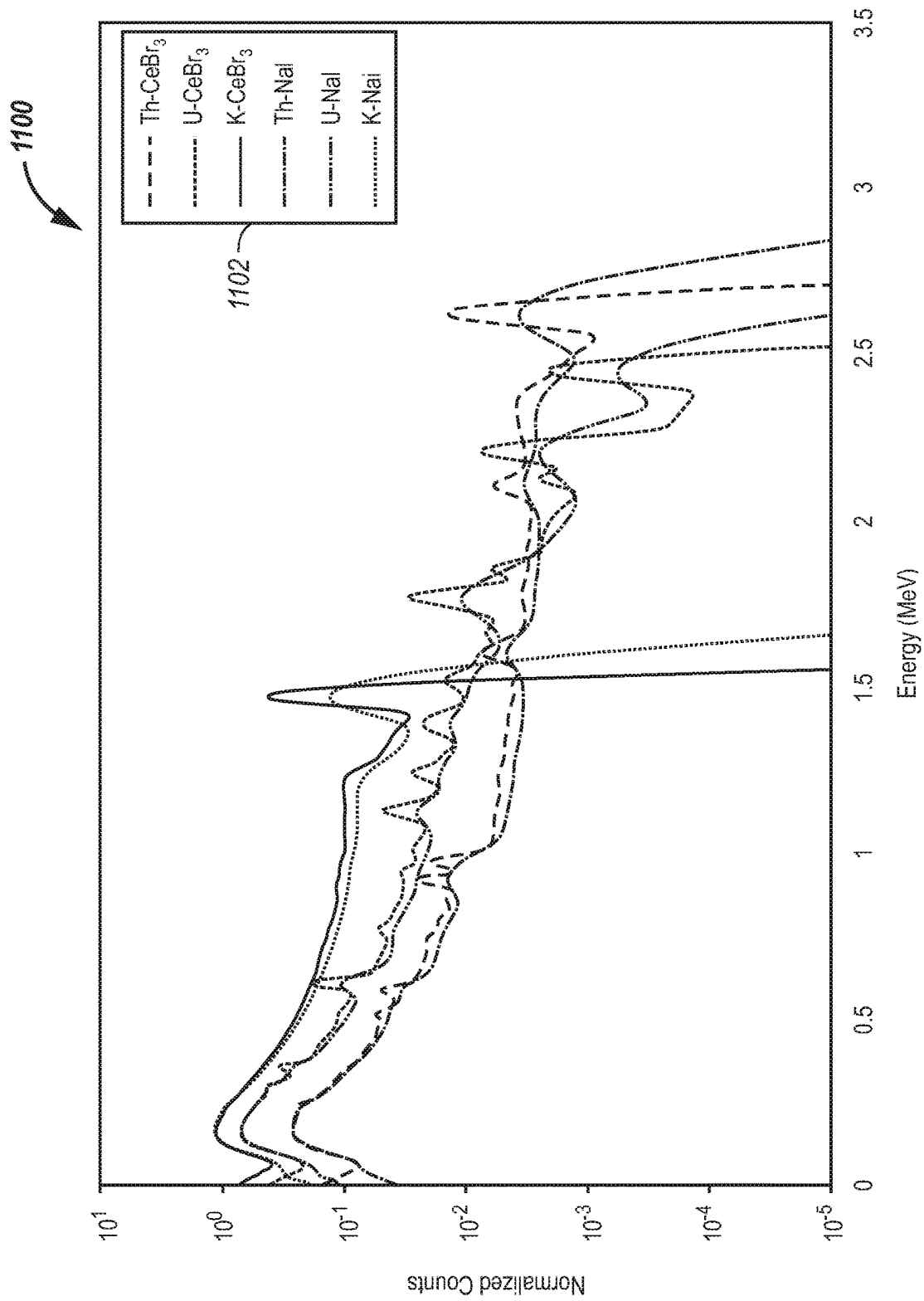


Figure 11

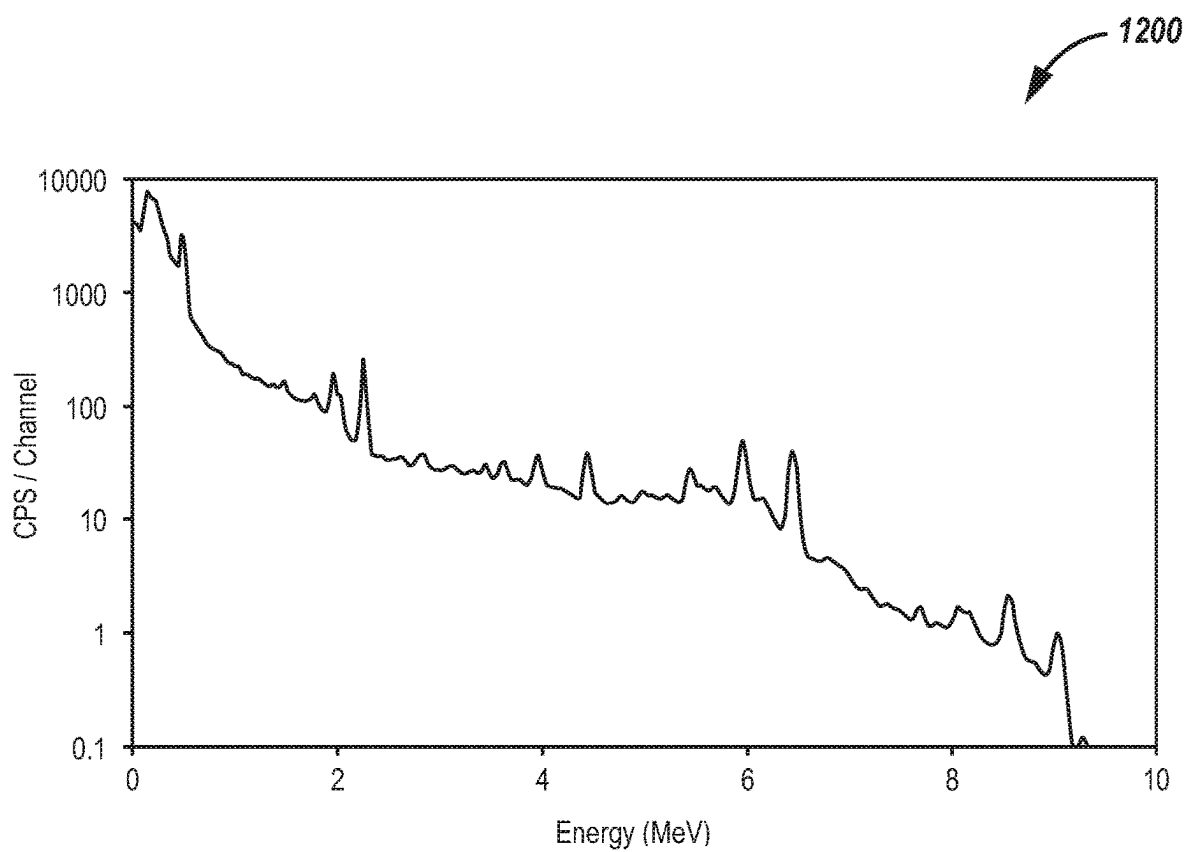


Figure 12

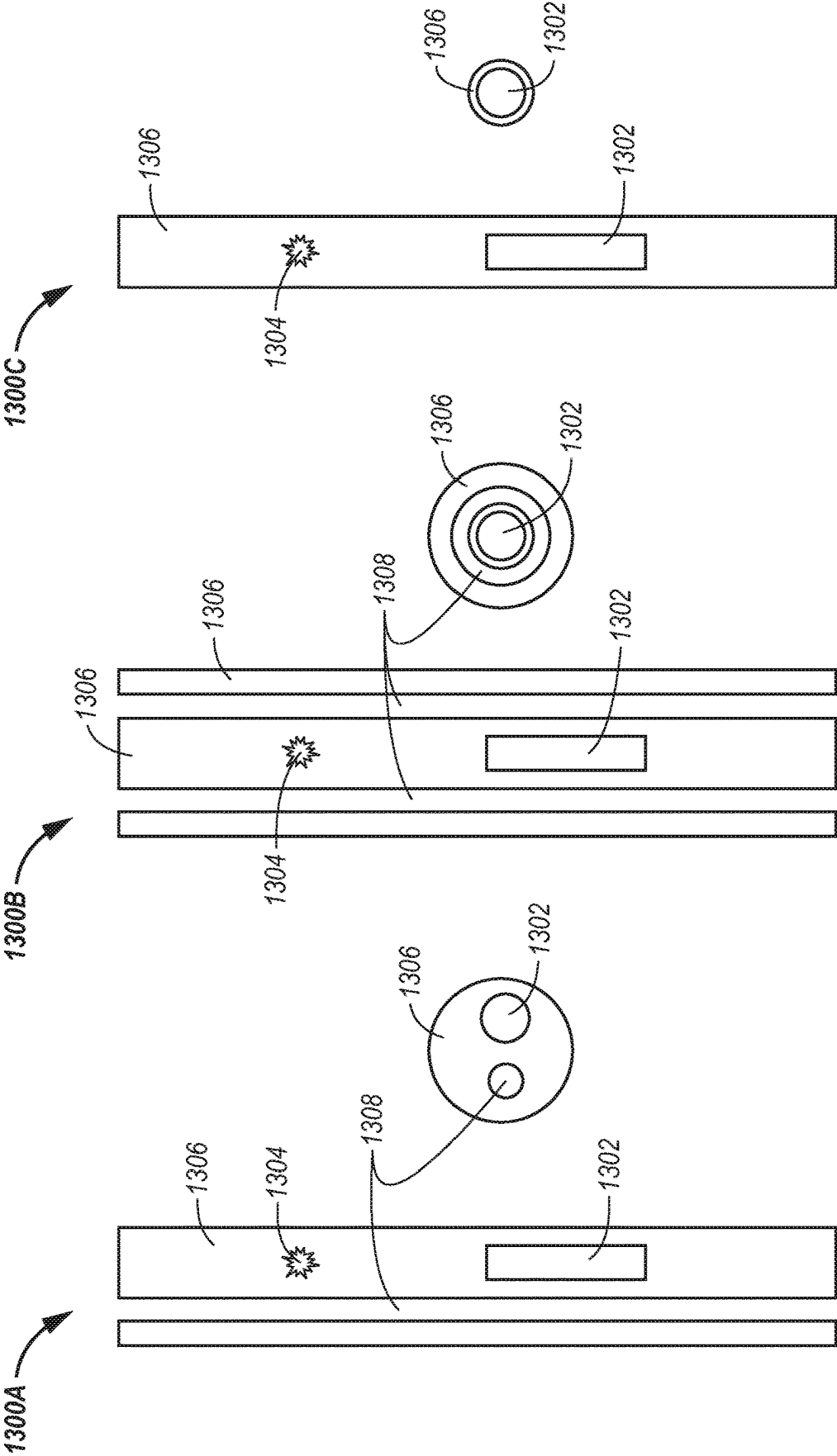


Figure 13C

Figure 13B

Figure 13A

GAMMA-RAY SPECTROSCOPY LOGGING TOOL

BACKGROUND

[0001] Wellbores drilled into subterranean formations may enable recovery of desirable fluids (e.g., hydrocarbons) using a number of different techniques. During drilling operations, slickline operations, or during wireline operations, measurements may be taken to determine the presence of oil, water, gas, and/or the like. One such device that may be utilized for these measurements may be a pulsed neutron tool. The pulsed neutron tool may comprise a pulsed neutron generator (PNG) that may operate and function to transmit neutrons into a formation for either logging while drilling (LWD) or measurement while drilling (MWD), or wireline logging measurements. In this context, “logging” may refer to collecting, recording, storing, and transmitting information.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] These drawings illustrate certain aspects of some of the embodiments of the present disclosure and should not be used to limit or define the disclosure.

[0003] FIG. 1 illustrates a logging while drilling operation utilizing a pulsed neutron logging tool, in accordance with examples of the present disclosure.

[0004] FIG. 2 illustrates the pulsed neutron logging tool in a wireline operation, in accordance with examples of the present disclosure.

[0005] FIG. 3 is a diagram of illustrative embodiments of a pulsed neutron logging tool.

[0006] FIG. 4A-4C are diagrams of alternative embodiments of a pulsed neutron logging tool.

[0007] FIG. 5 illustrates the energy of a neutron as it interacts in the present disclosure.

[0008] FIG. 6 illustrates a schematic of an information handling system.

[0009] FIG. 7 illustrates a schematic of a chip set.

[0010] FIG. 8 illustrates a computing network.

[0011] FIG. 9 illustrates a neural network.

[0012] FIG. 10 is a plot of intrinsic background spectra of different scintillator materials.

[0013] FIG. 11 is a plot of natural gamma-ray spectra of sodium iodide and cerium bromide (CeBr_3).

[0014] FIG. 12 is a plot of a neutron induced capture gamma-ray spectrum from a logging while drilling (LWD) tool equipped with CeBr_3 as a scintillator.

[0015] FIGS. 13A-C are diagrams of a pulsed neutron logging tool having a gamma ray scintillator detector with CeBr_3 as a scintillator for LWD and wireline operations.

DETAILED DESCRIPTION

[0016] The present disclosure generally relates to systems and methods for gamma-ray spectroscopy via a downhole logging tool. The logging tool may have, for example, a scintillator that is or includes cerium bromide (CeBr_3). Some aspects are directed to dual purpose gamma-ray spectroscopy via the logging tool in that implementations of the logging tool can detect or measure both neutron-induced gamma rays and natural gamma rays.

[0017] Embodiments include a pulsed neutron logging tool for a borehole in a subterranean formation. In examples, the logging tool includes a pulsed neutron generator to

broadcast neutrons into the subterranean formation and a gamma ray scintillator detector having a scintillator including CeBr_3 . The logging tool may be, for example, a logging while drilling (LWD) tool or a wireline tool. The pulsed neutron logging tool may be lowered into the borehole to detect (and measure), via the scintillator detector, natural gamma rays from the subterranean and neutron-induced gamma rays from the subterranean formation.

[0018] Neutron-induced gamma-ray spectroscopy and natural gamma-ray spectroscopy are measurements (e.g., logging) that provide information on the subterranean rock formation for oil and gas exploration and production. However, these two forms of spectroscopic measurements (one based on neutron-induced gamma rays and the other based on natural gamma rays) can conventionally generally only be achieved by two different respective tools (measurement systems) and thus typically in more than one pass or run in the wellbore, no matter if in wireline logging or while drilling applications. This conventional need for two different tools in two respective runs can be primarily due to the limitation of conventional gamma-ray detectors. Each conventional detector (crystal) has its own drawbacks, which make the optimization of such dual applications generally not feasible. In contrast, embodiments herein address this challenge in introducing CeBr_3 as a scintillator for a tool detector (e.g., of a neutron logging tool) utilized to measure both neutron-induced gamma rays and natural gamma rays, and having high resolution and low background in implementations.

[0019] The gamma ray detector can be a core technology of nuclear logging products and services in the well logging industry. Higher detected gamma-ray counting rate, better-characterized energy peaks, and better thermal stability may provide more accurate information about the surrounding rock formation while being constrained by the limited space in a logging or drilling tool and being challenged by the hostile downhole environments.

[0020] The compound CeBr_3 as an innovative material for a scintillator for gamma ray spectroscopy in accordance with embodiments herein can be an improvement over commonly used scintillators in logging measurements. The material CeBr_3 with its high density can give high light yield, fast decay time, good resolution, elevated temperature stability, and little or no intrinsic radiation background. Such can make CeBr_3 advantageous as a scintillator detector for nuclear logging products. The material CeBr_3 can be beneficial for new logging tools for gamma ray detection and spectroscopy and/or as an upgrade for existing logging tools for gamma ray detection and spectroscopy.

[0021] The logging tool can be a pulsed neutron logging tool for a borehole in a subterranean formation and that broadcasts (generates and emits) neutrons (e.g., via a pulsed neutron generator) into the subterranean formation. In implementations, the logging tool as a pulsed neutron logging tool equipped with CeBr_3 as the scintillator can be a dual-purpose gamma-ray spectroscopy logging tool. This pulsed neutron logging tool as a gamma-ray spectroscopy logging tool can be characterized as dual purpose in being operable in the borehole (e.g., on the same run into the borehole) to perform spectroscopic measurements of both (1) gamma rays from the subterranean formation induced by the broadcast of neutrons and (2) gamma rays naturally emitted from the subterranean formation (not induced by the broadcast of neutrons into the formation). In operation, the

pulsed neutron logging tool detects (measures) the neutron-induced gamma rays when the neutron generator (pulsed neutron generator) is turned ON and/or operating in the pulsing mode. In implementations, the pulsed neutron logging tool detects (measures) natural gamma rays when the neutron generator (pulsed neutron generator) is turned OFF (not the neutron OFF period during the pulsing mode, but the neutron generator is powered off or entirely not emitting neutrons for more than several milliseconds and most or all thermal neutrons being captured).

[0022] FIG. 1 is a diagram of an example drilling environment. Drilling environment **100** may include platform **102** that supports derrick **104** having a traveling block **108** for raising and lowering top drive **110** and drillstring **114**. Top drive **110** supports and rotates drillstring **114** as it is lowered through wellhead **112**. In turn, drill bit **124**, located at the end of drillstring **114**, may create borehole **116**. Borehole **116** may be formed through the Earth surface into a subterranean formation **126** in the Earth crust. Bottom-hole assembly **118** may include a pulsed neutron logging tool **132** (e.g., having a scintillator that is CeBr_3) for logging while drilling operations. Each of these components is described below. Pulsed neutron logging tool **132** may be a dual-purpose (dual application) gamma-ray spectroscopy logging tool in contemporaneously (e.g., on the same run) detecting (facilitating measuring) both (1) neutron-induced gamma rays from the subterranean formation **126** and (2) natural gamma rays from the subterranean formation **126**. In implementations for logging while drilling, such dual application can reduce complexity of bottom-hole assembly **118** and save rig time in facilitating spectroscopic measurements of both neutron-induced gamma rays and natural gamma rays in a single run (in the same run) into borehole **116**.

[0023] Platform **102** is a structure which may be used to support one or more other components of drilling environment **100** (e.g., derrick **104**). Platform **102** may be designed and constructed from suitable materials (e.g., concrete) which are able to withstand the forces applied by other components (e.g., the weight and counterforces experienced by derrick **104**). In any embodiment, platform **102** may be constructed to provide a uniform surface for drilling operations in drilling environment **100**.

[0024] Derrick **104** is a structure which may support, contain, and/or otherwise facilitate the operation of one or more pieces of the drilling equipment. In any embodiment, derrick **104** may provide support for crown block **106**, traveling block **108**, and/or any part connected to (and including) drillstring **114**. Derrick **104** may be constructed from any suitable materials (e.g., steel) to provide the strength necessary to support those components.

[0025] Crown block **106** is one or more simple machine(s) which may be rigidly affixed to derrick **104** and include a set of pulleys (e.g., a “block”), threaded (e.g., “reeved”) with a drilling line (e.g., a steel cable), to provide mechanical advantage. Crown block **106** may be disposed vertically above traveling block **108**, where traveling block **108** is threaded with the same drilling line.

[0026] Traveling block **108** is one or more simple machine(s) which may be movably affixed to derrick **104** and include a set of pulleys, threaded with a drilling line, to provide mechanical advantage. Traveling block **108** may be disposed vertically below crown block **106**, where crown block **106** is threaded with the same drilling line. In any embodiment, traveling block **108** may be mechanically coupled to drill-

string **114** (e.g., via top drive **110**) and allow for drillstring **114** (and/or any component thereof) to be lifted from (and out of) borehole **116**. Both crown block **106** and traveling block **108** may use a series of parallel pulleys (e.g., in a “block and tackle” arrangement) to achieve significant mechanical advantage, allowing for the drillstring to handle greater loads (compared to a configuration that uses non-parallel tension). Traveling block **108** may move vertically (e.g., up, down) within derrick **104** via the extension and retraction of the drilling line.

[0027] Top drive **110** is a machine which may be configured to rotate drillstring **114**. Top drive **110** may be affixed to traveling block **108** and configured to move vertically within derrick **104** (e.g., along with traveling block **108**). In any embodiment, the rotation of drillstring **114** (caused by top drive **110**) may allow for drillstring **114** to carve borehole **116**. Top drive **110** may use one or more motor(s) and gearing mechanism(s) to cause rotations of drillstring **114**. In any embodiment, a rotatory table (not shown) and a “Kelly” drive (not shown) may be used in addition to, or instead of, top drive **110**.

[0028] Wellhead **112** is a machine which may include one or more pipes, caps, and/or valves to provide pressure control for contents within borehole **116** (e.g., when fluidly connected to a well (not shown)). In any embodiment, during drilling, wellhead **112** may be equipped with a blowout preventer (not shown) to prevent the flow of higher-pressure fluids (in borehole **116**) from escaping to the surface in an uncontrolled manner. Wellhead **112** may be equipped with other ports and/or sensors to monitor pressures within borehole **116** and/or otherwise facilitate drilling operations.

[0029] Drillstring **114** is a machine which may be used to carve borehole **116** and/or gather data from borehole **116** and the surrounding geology. Drillstring **114** may include one or more drillpipe(s), one or more repeater(s) **122**, and bottom-hole assembly **118**. Drillstring **114** may rotate (e.g., via top drive **110**) to form and deepen borehole **116** (e.g., via drill bit **124**) and/or via one or more motor(s) attached to drillstring **114**.

[0030] Borehole **116** is a hole in the ground which may be formed by drillstring **114** (and one or more components thereof). Borehole **116** may be partially or fully lined with casing to protect the surrounding ground from the contents of borehole **116**, and conversely, to protect borehole **116** from the surrounding ground.

[0031] Bottom-hole assembly **118** is a machine which may be equipped with one or more tools for creating, providing structure, and maintaining borehole **116**, as well as one or more tools for measuring the surrounding environment (e.g., measurement while drilling (MWD), logging while drilling (LWD)). In any embodiment, bottom-hole assembly **118** may be disposed at (or near) the end of drillstring **114** (e.g., in the most “downhole” portion of borehole **116**).

[0032] Non-limiting examples of tools that may be included in bottom-hole assembly **118** include a drill bit (e.g., drill bit **124**), casing tools (e.g., a shifting tool), a plugging tool, a mud motor, a drill collar (thick-walled steel pipes that provide weight and rigidity to aid the drilling process), actuators (and pistons attached thereto), a steering system, and any measurement tool (e.g., sensors, probes, particle generators, etc.).

[0033] Further, bottom-hole assembly **118** may include a telemetry sub to maintain a communications link with the

surface (e.g., with information handling system 120). Such telemetry communications may be used for (i) transferring tool measurement data from bottom-hole assembly 118 to surface receivers, and/or (ii) receiving commands (from the surface) to bottom-hole assembly 118 (e.g., for use of one or more tool(s) in bottom-hole assembly 118). In examples, telemetry communications may be at least in part between bottom-hole assembly 118 and information handling system 120.

[0034] As illustrated, the information handling system 120 may comprise any instrumentality or aggregate of instrumentalities operable to compute, estimate, classify, process, transmit, broadcast, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system 120 may be a personal computer, a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price.

[0035] Information handling system 120 may include a processing unit (e.g., microprocessor, central processing unit, etc.) that may process drilling data from rotary steerable system (RSS) 242, discussed below, by executing software or instructions obtained from a local non-transitory computer readable media (e.g., optical disks, magnetic disks). The non-transitory computer readable media may store software or instructions of the methods described herein. Non-transitory computer readable media may include any instrumentality or aggregation of instrumentalities that may retain data and/or instructions for a period of time. Non-transitory computer readable media may include, for example, storage media such as a direct access storage device (e.g., a hard disk drive or floppy disk drive), a sequential access storage device (e.g., a tape disk drive), compact disk, CD-ROM, DVD, random access memory (RAM), read only memory (ROM), electrically erasable programmable read-only memory (EEPROM), and/or flash memory; as well as communications media such wires, optical fibers, microwaves, radio waves, and other electromagnetic and/or optical carriers; and/or any combination of the foregoing. Information handling system 120 may also include input device(s) (e.g., keyboard, mouse, touchpad, etc.) and output device(s) (e.g., monitor, printer, etc.). The input device(s) and output device(s) provide a user interface that enables an operator to interact with any device disposed or a part of bottom-hole assembly 118, discussed below, and/or software executed by a processing unit. For example, information handling system 120 may enable an operator to select analysis options, view collected log data, view analysis results, and/or perform other tasks.

[0036] Non-limiting examples of techniques for transferring tool measurement data (to the surface) include mud pulse telemetry and through-wall acoustic signaling. For through-wall acoustic signaling, one or more repeater(s) 122 may detect, amplify, and re-transmit signals from bottom-hole assembly 118 to the surface (e.g., to information handling system 120), and conversely, from the surface (e.g., from information handling system 120) to bottom-hole assembly 118.

[0037] Repeater 122 is a device which may be used to receive and send signals from one component of drilling environment 100 to another component of drilling environment 100. As a non-limiting example, repeater 122 may be

used to receive a signal from a tool on bottom-hole assembly 118 and send that signal to information handling system 120. Two or more repeaters 122 may be used together, in series, such that a signal to/from bottom-hole assembly 118 may be relayed through two or more repeaters 122 before reaching its destination.

[0038] A transducer is a device that may work with repeater 122 to transfer information from the surface to bottom-hole assembly 118. A transducer may be configured to convert non-digital data (e.g., vibrations, other analog data) into a digital form suitable for information handling system 120. As a non-limiting example, the one or more transducer(s) may convert signals between mechanical and electrical forms, enabling information handling system 120 to receive the signals from a telemetry sub, on bottom-hole assembly 118, and conversely, transmit a downlink signal to the telemetry sub on bottom-hole assembly 118. In any embodiment, the transducer may be located at the surface and/or any part of drillstring 114 (e.g., as part of bottom-hole assembly 118).

[0039] Drill bit 124 is a machine which may be used to cut through, scrape, and/or crush (i.e., break apart) materials in the ground (e.g., rocks, dirt, clay, etc.). Drill bit 124 may be disposed at the frontmost point of drillstring 114 and bottom-hole assembly 118. In any embodiment, drill bit 124 may include one or more cutting edges (e.g., hardened metal points, surfaces, blades, protrusions, etc.) to form a geometry which aids in breaking ground materials loose and further crushing that material into smaller sizes. In any embodiment, drill bit 124 may be rotated and forced into (i.e., pushed against) the ground material to cause the cutting, scraping, and crushing action. The rotations of drill bit 124 may be caused by top drive 110 and/or one or more motor(s) located on drillstring 114 (e.g., on bottom-hole assembly 118).

[0040] Pump 128 is a machine that may be used to circulate drilling fluid 130 from a reservoir, through a feed pipe, to derrick 104, to the interior of drillstring 114, out through drill bit 124 (through orifices, not shown), back upward through borehole 116 (around drillstring 114), and back into the reservoir. In any embodiment, any appropriate pump 128 may be used (e.g., centrifugal, gear, etc.) which is powered by any suitable means (e.g., electricity, combustible fuel, etc.).

[0041] Drilling fluid 130 is a liquid which may be pumped through drillstring 114 and borehole 116 to collect drill cuttings, debris, and/or other ground material from the end of borehole 116 (e.g., the volume most recently hollowed by drill bit 124). Further, drilling fluid 130 may provide conductive cooling to drill bit 124 (and/or bottom-hole assembly 118). In any embodiment, drilling fluid 130 may be circulated via pump 128 and filtered to remove unwanted debris.

[0042] During drilling operations, bottom-hole assembly may comprise, at least in part, a pulsed neutron logging tool 132. This may allow for logging while drilling operations to be performed. Measurements taken by pulsed neutron logging tool 132 may be gathered and/or processed by information handling system 120. For example, measurements taken by pulsed neutron logging tool 132 may be sent to information handling system 120 where they may be stored on memory and then processed. The processing may be performed real-time during data acquisition or after recovery of pulsed neutron logging tool 132. Processing may alter-

natively occur downhole on an information handling system disposed on and/or near pulsed neutron logging tool 132 or may occur both downhole and at surface. Information handling system 120 may process the signals, and the information contained therein may be displayed for an operator to observe and stored for future processing and reference. Information handling system 120 may also contain an apparatus for supplying control signals and power to pulsed neutron logging tool 132. Although illustrated as disposed on bottom-hole assembly 118 in a drilling operation, pulsed neutron logging tool 132 may also be disposed in borehole 116 in a wireline operation. Moreover, as mentioned, pulsed neutron logging tool 132 can have a scintillator detector having a scintillator (scintillation crystal) that is or includes CeBr_3 .

[0043] FIG. 2 illustrates a wireline operation 200, as disclosed herein, utilizing a pulsed neutron logging tool 132. Pulsed neutron logging tool 132 can have a scintillator detector in which the scintillator may be or include CeBr_3 . FIG. 2 illustrates a cross-section of borehole 116 with a pulsed neutron logging tool 132 traveling through casing string 202. Borehole 116 may traverse through subterranean formation 204 as a vertical well and/or a horizontal well. Pulsed neutron logging tool 132 may be suspended by a conveyance 206, which communicates power from a logging facility 216 (logging center) to pulsed neutron logging tool 132 and communicates telemetry from pulsed neutron logging tool 132 to information handling system 120. In examples, pulsed neutron logging tool 132 may be operatively coupled to a conveyance 206 (e.g., wireline, slickline, coiled tubing, pipe, downhole tractor, and/or the like) which may provide mechanical suspension, as well as electrical connectivity, for pulsed neutron logging tool 132. Conveyance 206 and pulsed neutron logging tool 132 may extend within casing string 202 to a depth within borehole 116. Conveyance 206, which may include one or more electrical conductors, may exit wellhead 112, may pass around pulley 208, may engage odometer 210, and may be reeled onto winch 212, which may be employed to raise and lower the tool assembly in borehole 116. Wellhead 112 may allow for entry into borehole 116 and placement of pulsed neutron logging tool 132 into pipe string 214. The position of pulsed neutron logging tool 132 may be monitored in a number of ways, including an inertial tracker in pulsed neutron logging tool 132 and a paid-out conveyance length monitor in logging facility 216.

[0044] Multiple such measurements may be desirable to enable the system to compensate for varying cable tension and cable stretch due to other factors. Information handling system 120 in logging facility 216 collects telemetry and position measurements and provides position-dependent logs of measurements from pulsed neutron logging tool 132 and values that may be derived therefrom.

[0045] Pulsed neutron logging tool 132 generally includes multiple instruments for measuring a variety of downhole parameters. Wheels, bow springs, fins, pads, or other centralizing mechanisms may be employed to keep pulsed neutron logging tool 132 near the borehole axis during measurement operations. During measurement operations, generally, measurements may be performed as pulsed neutron logging tool 132 is drawn up hole at a constant rate. The parameters and instruments may vary depending on the needs of the measurement operation.

[0046] Measurements taken by pulsed neutron logging tool 132 may be gathered and/or processed by information handling system 120. For example, signals recorded by pulsed neutron logging tool 132 may be sent to information handling system 120 where they may be stored on memory and then processed. The processing may be performed real-time during data acquisition or after recovery of pulsed neutron logging tool 132. Processing may alternatively occur downhole on an information handling system disposed on pulsed neutron logging tool 132 or may occur both downhole and at surface. In some examples, signals recorded by pulsed neutron logging tool 132 may be conducted to information handling system 120 by way of conveyance 206. Information handling system 120 may process the signals, and the information contained therein may be displayed for an operator to observe and stored for future processing and reference. Information handling system 120 may also contain an apparatus for supplying control signals and power to pulsed neutron logging tool 132.

[0047] In wireline operations 200, a digital telemetry system may be employed, wherein an electrical circuit may be used to both supply power to pulsed neutron logging tool 132 and to transfer data between information handling system 120 and pulsed neutron logging tool 132. A DC voltage may be provided to pulsed neutron logging tool 132 by a power supply located above ground level, and data may be coupled to the DC power conductor by a baseband current pulse system. Alternatively, pulsed neutron logging tool 132 may be powered by batteries located within the downhole tool assembly, and/or the data provided by pulsed neutron logging tool 132 may be stored within the downhole tool assembly, rather than transmitted to the surface during logging.

[0048] FIG. 3 illustrates pulsed neutron logging tool 132 disposed in borehole 116. It should be noted, as discussed above, that pulsed neutron logging tool 132 may be disposed on a bottom-hole assembly 118 (e.g., referring to FIG. 1) in a logging while drilling operation or utilized in a wireline operation (e.g., referring to FIG. 2). Additionally, the orientation of pulsed neutron logging tool 132, whether the pulsed neutron generator is disposed above or below the detectors, is inconsequential.

[0049] With continued reference to FIG. 3, pulsed neutron logging tool 132 may comprise an outer housing 300 which may be formed from a heavy metal such as steel, Inconel, etc. Housing 300 may protect the internal devices of pulsed neutron logging tool 132 from the downhole environment that pulsed neutron logging tool 132 may experience in borehole 116. As illustrated, pulsed neutron logging tool 132 may be divided into a generator area 302 (generation area) and a detection area 304 (detector area) that are separated by shielding 306. From generator area 302, neutrons may be generated and broadcast into formation 204 (referring to FIG. 2). Detection area 304 may be operated and function to detect gamma rays that may originate from formation 204 naturally or induced by the broadcast of neutrons into formation 204.

[0050] Generator area 302 may comprise a pulsed neutron generator 308 that may be packaged within silicon hexafluoride (SF_6) housing 310 that has or is configured for SF_6 gas therein as an insulating gas for a high voltage environment. SF_6 housing 310 (an inner housing within outer housing 300) may be comprised of a heavy metal like stainless steel, etc. As noted above, within SF_6 housing 310 may be a pulsed

neutron generator **308** that may further comprise a neutron tube **312**, which generates neutrons for broadcasting, and a high voltage (HV) ladder power supply **314** that may be utilized to power neutron tube **312**. In other examples, pulsed neutron generator **308** may be replaced with a continuous neutron source such as Americium-Beryllium (Am—Be) chemical source or other types of chemical sources. Moreover, while housing **310** is labeled as an SF₆ housing, the housing **310** may house (contain) another insulating gas (dielectric gas) in lieu of SF₆. Further, liquid or solid media (instead of an insulating gas) may be employed as an insulator of the high voltage. Outside of SF₆ housing **310** may be a fast neutron monitor **316**, that may be utilized to monitor the broadcasting of neutrons **318** from generator area **302** into formation **204**. For example, during operations, pulsed neutron logging tool **132** may generate pulses of high energy neutrons that radiate from pulsed neutron generator **308** into the surrounding environment including borehole **116** and formation **204**. The highly energetic neutrons **318** entering the surrounding environment interact with atomic nuclei, inducing gamma ray radiation. Induced inelastic and capture gamma rays **320** and thermal neutrons **328** may be sensed and recorded by detection area **304**. The inelastic and capture gamma rays **320** sensed and recorded include inelastic gamma rays. The inelastic and capture gamma rays **320** sensed and recorded include capture gamma rays different than the inelastic gamma rays. The scattered neutrons and gamma ray spectrum may be measured to determine properties of borehole **116** and formation **204**. Through processing, the measurements may be utilized to identify oil and gas in formation **204** as well as determining the flow in production wells. As illustrated, neutrons **318** may be broadcasted into formation **204**, wherein neutrons **318** may interact with material within formation **204** to create inelastic and capture gamma rays **320**, discussed in greater detail below. Inelastic and capture gamma rays **320** may be detected, sensed, and/or measured by devices within detection area **304** of pulsed neutron logging tool **132**. Inelastic gamma rays are generally induced by fast neutrons while capture gamma rays are induced by thermal neutrons capturing. Natural gamma rays naturally occur from the formation **204** and can be measured via the pulsed neutron logging tool **132** with the pulsed neutron generator **308** turned OFF (not generating or pulsing neutrons), or with the pulsed neutron generator **308** in a prolonged OFF state in a given pulsing sequence. The technique may include performing in a borehole, via the pulsed neutron logging tool, spectroscopy measurements of natural gamma rays, pulsed neutron induced inelastic gamma rays, and pulsed neutron induced capture gamma rays.

[0051] Detection area **304** may comprise a number of devices that may be utilized to detect, sense, and/or measure inelastic and capture gamma rays **320**. As illustrated, a number of gamma ray scintillator detectors may be utilized, which implement a scintillation crystal (e.g., CeBr₃) coupled to a photomultiplier tube. The scintillation crystal may be labeled as a scintillator **327**. In examples, gamma ray scintillator detectors may be identified as a near gamma ray scintillator detector **322**, a far gamma ray scintillator detector **324**, and a long gamma ray scintillator detector **326**. Identification of each scintillator detector as near, far, and long is due to the distance from neutron generator **308**. For example, the closest scintillator detector to neutron genera-

tor **308** is “near,” the second closest is “far”, and the third closest is “long.” This nomenclature may also be utilized for thermal neutron detectors that may also be disposed within detection area **304** and may operate and function to detect thermal neutrons **328** that may originate from formation **204** during the interaction of neutrons **318** with material within formation **204**. For example, neutron detectors may operate and function to count thermal (around about 0.025 eV) and/or epithermal (between about 0.1 eV and 100 eV) neutrons. Suitable neutron detectors include Helium-3 (He-3) filled proportional counters, though other neutron counters may also be used. Thus, within detection area **304** may be a near thermal neutron detector **330**, a far thermal neutron detector **332**, and a long thermal neutron detector **334**. As noted above, detection area **304** may be separated from generator area **302** by shielding **306**.

[0052] Shielding **306** may be a structure formed of a heavy metal like tungsten. This material may operate and function to prevent neutrons **318** that may be generated from pulsed neutron generator **308** from being detected by the detectors in detection area **304**. Without shielding **306**, neutrons **318** generated from pulsed neutron generator **308** may saturate all detectors within detection area **304** and prevent the detection and measurement of gamma rays and neutrons from formation **204**.

[0053] FIGS. 4A-4C illustrate different embodiments of pulsed neutron logging tool **132**. FIG. 4A illustrates an embodiment shown in FIG. 3. In this embodiment, the distance from pulsed neutron generator **308** to near thermal neutron detector **330** is D_{n1}, to far thermal neutron detector **332** is D_{n2}, and to long thermal neutron detector **334** is D_{n3}. Further, the distance from pulsed neutron generator **308** to a near gamma ray scintillator detector **322** is D_{γ1}, a far gamma ray scintillator detector **324** is D_{γ2}, and a long gamma ray scintillator detector **326** is D_{γ3}. FIG. 4B illustrates another embodiment in which the distances D_{n1}, D_{n2}, D_{n3} from pulsed neutron generator **308** to each thermal neutron detector **330**, **332**, **334** have changed as each thermal neutron detector is now disposed within generator area **302**. FIG. 4C illustrates an embodiment where gamma ray scintillator detectors **332**, **324**, and **326** (with distances D_{γ1}, D_{γ2}, D_{γ3}) are utilized, but thermal neutron detectors **330**, **332**, **334** are not utilized.

[0054] Multiple detectors of pulsed neutron logging tool **132**, may enable pulsed neutron logging tool **132** to measure properties of formation **204** and borehole **116** (e.g., referring to FIG. 3) using any of the existing multiple-spacing techniques. In addition, the presence of gamma ray detectors (e.g., **320**, **322**, **324**) which have proper distances from pulsed neutron generator **308**, may enable the measurement of elemental gamma ray spectroscopy.

[0055] As discussed above, during measurement operations, neutrons **318** (e.g., referring to FIG. 3) emitted from neutron source or pulsed neutron generator **308** undergo neutron scattering and/or nuclear absorption when interacting with matter. Scattering may either be elastic (n, n) or inelastic (n, n'). In an elastic interaction a fraction of the neutrons kinetic energy is transferred to the nucleus. An inelastic interaction is similar, except the nucleus undergoes an internal rearrangement. Additionally, neutrons may also undergo an absorption interaction. During interactions, the elastic cross section is nearly constant, whereas the inelastic scattering cross section and absorption cross sections are proportional to the reciprocal of the neutron speed. For

example, inelastic scatterings appear for fast neutrons in the MeV energy range, whereas absorptions happen when neutrons slowed down in the eV energy range.

[0056] FIG. 5 illustrates a graph 500 that depicts different scattering by a neutron 318. As illustrated, neutron 318 may be traveling at a fast speed with high kinetic energy and interacts with nuclei 504, releasing inelastic gamma ray 320 and lowering the energy state of neutron 318. After the interaction, neutron 318 contains too much energy to be absorbed, thus continuing its path until it interacts with nuclei 508 releasing inelastic gamma ray 320 and again lowering its energy state again. After the interactions, neutron 318 has kinetic energy close to target energy 512, becomes a thermal neutron 328. Thus, when neutron 328 at target energy 512 interacts with nuclei 514 it will be captured. This interaction results in nucleus 514 being rearranged to contain previously traveling neutron 328 and an emitted capture gamma ray 320. Sensing these events with pulsed neutron logging tool 132 using detection area 304 may allow for the identification of oil, gas, and/or water in borehole 116 and formation 204 (e.g., referring to FIG. 3).

[0057] With continued reference to FIG. 5, the neutron to gamma ray timing information may be utilized during measurement operations in which a pulsing neutron generator is utilized. In a sub-us time domain, inelastic gamma rays dominate, whereas in a 10-1000 us time range, there are only capture gamma rays. Insert 520 on FIG. 5 illustrates an example of neutrons in a neutron pulse 522 and insert 524 shows the relationship of two adjacent neutron pulses 522 with a given pulse width and timing interval. Pulsing schemes allow isolation of inelastic and capture gamma rays 320, and then allow elemental determinations of different nuclei in the bore hole, formation, or fluids.

[0058] During measurement operations, pulsed neutron logging tool 132 may take any number of measurements of inelastic and capture gamma rays 320 and/or thermal neutrons 328 (e.g., referring to FIG. 3). The pulsed neutron logging tool 132 may also take measurements of natural gamma rays from the formation 204 with the pulsed neutron generator 308 in a prolonged OFF state (not generating neutrons) with a given pulsing sequence. These measurements may be further processed by additional methods and systems that may utilize information handling system 120.

[0059] FIG. 6 further illustrates an example information handling system 120 which may be employed to perform various steps, methods, and techniques disclosed herein. Persons of ordinary skill in the art will readily appreciate that other system examples are possible. As illustrated, information handling system 120 includes a processing unit (CPU or processor) 602 and a system bus 604 that couples various system components including system memory 606 such as ROM 608 and RAM 610 to processor 602. Processors disclosed herein may all be forms of this processor 602. Information handling system 120 may include a cache 612 of high-speed memory connected directly with, in close proximity to, or integrated as part of processor 602. Information handling system 120 copies data from memory 606 and/or storage device 614 to cache 612 for quick access by processor 602. In this way, cache 612 provides a performance boost that avoids processor 602 delays while waiting for data. These and other modules may control or be configured to control processor 602 to perform various operations or actions. Other system memory 606 may be available for use as well. Memory 606 may include multiple

different types of memory with different performance characteristics. It may be appreciated that the disclosure may operate on information handling system 120 with more than one processor 602 or on a group or cluster of computing devices networked together to provide greater processing capability. Processor 602 may include any general-purpose processor and a hardware module or software module, such as first module 616, second module 618, and third module 620 stored in storage device 614, configured to control processor 602 as well as a special-purpose processor where software instructions are incorporated into processor 602. Processor 602 may be a self-contained computing system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric. Processor 602 may include multiple processors, such as a system having multiple, physically separate processors in different sockets, or a system having multiple processor cores on a single physical chip. Similarly, processor 602 may include multiple distributed processors located in multiple separate computing devices but working together such as via a communications network. Multiple processors or processor cores may share resources such as memory 606 or cache 612 or may operate using independent resources. Processor 602 may include one or more state machines, an application specific integrated circuit (ASIC), or a programmable gate array (PGA) including a field PGA (FPGA).

[0060] Each individual component discussed above may be coupled to system bus 604, which may connect each and every individual component to each other. System bus 604 may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. A basic input/output (BIOS) stored in ROM 608 or the like, may provide the basic routine that helps to transfer information between elements within information handling system 120, such as during start-up. Information handling system 120 further includes storage devices 614 or computer-readable storage media such as a hard disk drive, a magnetic disk drive, an optical disk drive, tape drive, solid-state drive, RAM drive, removable storage devices, a redundant array of inexpensive disks (RAID), hybrid storage device, or the like. Storage device 614 may include software modules 616, 618, and 620 for controlling processor 602. Information handling system 120 may include other hardware or software modules. Storage device 614 is connected to the system bus 604 by a drive interface. The drives and the associated computer-readable storage devices provide nonvolatile storage of computer-readable instructions, data structures, program modules and other data for information handling system 120. In one aspect, a hardware module that performs a particular function includes the software component stored in a tangible computer-readable storage device in connection with hardware components, such as processor 602, system bus 604, and so forth, to carry out a particular function. In another aspect, the system may use a processor and computer-readable storage device to store instructions which, when executed by the processor, cause the processor to perform operations, a method or other specific actions. The basic components and appropriate variations may be modified depending on the type of device, such as whether information handling system 120 is a small, handheld computing device, a desktop computer, or a computer server. When processor 602 executes instructions to perform

“operations”, processor 602 may perform the operations directly and/or facilitate, direct, or cooperate with another device or component to perform the operations.

[0061] As illustrated, information handling system 120 employs storage device 614, which may be a hard disk or other types of computer-readable storage devices which may store data that are accessible by a computer, such as magnetic cassettes, flash memory cards, digital versatile disks (DVDs), cartridges, random access memories (RAMs) 610, read only memory (ROM) 608, a cable containing a bit stream and the like, may also be used in the exemplary operating environment. Tangible computer-readable storage media, computer-readable storage devices, or computer-readable memory devices, expressly exclude media such as transitory waves, energy, carrier signals, electromagnetic waves, and signals per se.

[0062] To enable user interaction with information handling system 120, an input device 622 represents any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech and so forth. Additionally, input device 622 may receive one or more measurements from bottom-hole assembly 118 (e.g., referring to FIG. 1), discussed above. An output device 624 may also be one or more of a number of output mechanisms known to those of skill in the art. In some instances, multimodal systems enable a user to provide multiple types of input to communicate with information handling system 120. Communications interface 626 generally governs and manages the user input and system output. There is no restriction on operating on any particular hardware arrangement and therefore the basic hardware depicted may easily be substituted for improved hardware or firmware arrangements as they are developed.

[0063] As illustrated, each individual component described above is depicted and disclosed as individual functional blocks. The functions these blocks represent may be provided through the use of either shared or dedicated hardware, including, but not limited to, hardware capable of executing software and hardware, such as a processor 602, that is purpose-built to operate as an equivalent to software executing on a general-purpose processor. For example, the functions of one or more processors presented in FIG. 6 may be provided by a single shared processor or multiple processors. (Use of the term “processor” should not be construed to refer exclusively to hardware capable of executing software.) Illustrative embodiments may include microprocessor and/or digital signal processor (DSP) hardware, read-only memory (ROM) 608 for storing software performing the operations described below, and random-access memory (RAM) 610 for storing results. Very large-scale integration (VLSI) hardware embodiments, as well as custom VLSI circuitry in combination with a general-purpose DSP circuit, may also be provided.

[0064] FIG. 7 illustrates an example information handling system 120 having a chipset architecture that may be used in executing the described method and generating and displaying a graphical user interface (GUI). Information handling system 120 is an example of computer hardware, software, and firmware that may be used to implement the disclosed technology. Information handling system 120 may include a processor 602, representative of any number of physically and/or logically distinct resources capable of executing software, firmware, and hardware configured to perform

identified computations. Processor 602 may communicate with a chipset 700 that may control input to and output from processor 602. In this example, chipset 700 outputs information to output device 624, such as a display, and may read and write information to storage device 614, which may include, for example, magnetic media, and solid-state media. Chipset 700 may also read data from and write data to RAM 610. A bridge 702 for interfacing with a variety of user interface components 704 may be provided for interfacing with chipset 700. Such user interface components 704 may include a keyboard, a microphone, touch detection and processing circuitry, a pointing device, such as a mouse, and so on. In general, inputs to information handling system 120 may come from any of a variety of sources, machine generated and/or human generated.

[0065] Chipset 700 may also interface with one or more communication interfaces 626 that may have different physical interfaces. Such communication interfaces may include interfaces for wired and wireless local area networks, for broadband wireless networks, as well as personal area networks. Some applications of the methods for generating, displaying, and using the GUI disclosed herein may include receiving ordered datasets over the physical interface or be generated by the machine itself by processor 602 analyzing data stored in storage device 614 or RAM 610. Further, information handling system 120 receives inputs from a user via user interface components 704 and executes appropriate functions, such as browsing functions by interpreting these inputs using processor 602.

[0066] In examples, information handling system 120 may also include tangible and/or non-transitory computer-readable storage devices for carrying or having computer-executable instructions or data structures stored thereon. Such tangible computer-readable storage devices may be any available device that may be accessed by a general purpose or special purpose computer, including the functional design of any special purpose processor as described above. By way of example, and not limitation, such tangible computer-readable devices may include RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other device which may be used to carry or store program code in the form of computer-executable instructions, data structures, or processor chip design. When information or instructions are provided via a network, or another communications connection (either hardwired, wireless, or combination thereof), to a computer, the computer properly views the connection as a computer-readable medium. Thus, any such connection is properly termed a computer-readable medium. Combinations of the above should also be included within the scope of the computer-readable storage devices.

[0067] Computer-executable instructions include, for example, instructions and data which cause a general-purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions. Computer-executable instructions also include program modules that are executed by computers in stand-alone or network environments. Generally, program modules include routines, programs, components, data structures, objects, and the functions inherent in the design of special-purpose processors, etc. that perform particular tasks or implement particular abstract data types. Computer-executable instructions, associated data structures, and program modules represent examples of the program code for

executing steps of the methods disclosed herein. The particular sequence of such executable instructions or associated data structures represents examples of corresponding acts for implementing the functions described in such steps.

[0068] In additional examples, methods may be practiced in network computing environments with many types of computer system configurations, including personal computers, hand-held devices, multi-processor systems, micro-processor-based or programmable consumer electronics, network PCs, minicomputers, mainframe computers, and the like. Examples may also be practiced in distributed computing environments where tasks are performed by local and remote processing devices that are linked (either by hard-wired links, wireless links, or by a combination thereof) through a communications network. In a distributed computing environment, program modules may be located in both local and remote memory storage devices.

[0069] FIG. 8 illustrates an example of one arrangement of resources in a computing network **800** that may employ the processes and techniques described herein, although many others are of course possible. As noted above, an information handling system **120**, as part of their function, may utilize data, which includes files, directories, metadata (e.g., access control list (ACLs) creation/edit dates associated with the data, etc.), and other data objects. The data on the information handling system **120** is typically a primary copy (e.g., a production copy). During a copy, backup, archive or other storage operation, information handling system **120** may send a copy of some data objects (or some components thereof) to a secondary storage computing device **804** by utilizing one or more data agents **802**.

[0070] A data agent **802** may be a desktop application, website application, or any software-based application that is run on information handling system **120**. As illustrated, information handling system **120** may be disposed at any rig site (e.g., referring to FIG. 1), off site location, or repair and manufacturing center. The data agent may communicate with a secondary storage computing device **804** using communication protocol **808** in a wired or wireless system. Communication protocol **808** may function and operate as an input to a website application. In the website application, field data related to pre- and post-operations, generated DTCs, notes, and the like may be uploaded. Additionally, information handling system **120** may utilize communication protocol **808** to access processed measurements, operations with similar DTCs, troubleshooting findings, historical run data, and/or the like. This information is accessed from secondary storage computing device **804** by data agent **802**, which is loaded on information handling system **120**.

[0071] Secondary storage computing device **804** may operate and function to create secondary copies of primary data objects (or some components thereof) in various cloud storage sites **806A-N**. Additionally, secondary storage computing device **804** may run determinative algorithms on data uploaded from one or more information handling systems **120**, discussed further below. Communications between the secondary storage computing devices **804** and cloud storage sites **806A-N** may utilize REST protocols (Representational state transfer interfaces) that satisfy basic C/R/U/D semantics (Create/Read/Update/Delete semantics), or other hypertext transfer protocol (“HTTP”)-based or file-transfer protocol (“FTP”)-based protocols (e.g., Simple Object Access Protocol).

[0072] In conjunction with creating secondary copies in cloud storage sites **806A-N**, the secondary storage computing device **804** may also perform local content indexing and/or local object-level, sub-object-level or block-level deduplication when performing storage operations involving various cloud storage sites **806A-N**. Cloud storage sites **806A-N** may further record and maintain, EM logs, map DTC codes, store repair and maintenance data, store operational data, and/or provide outputs from determinative algorithms that are located in cloud storage sites **806A-N**. In a non-limiting example, this type of network may be utilized as a platform to store, backup, analyze, import, preform extract, transform and load (“ETL”) processes, mathematically process, apply machine learning models, and augment EM measurement data sets.

[0073] A machine learning model may be an empirically derived model which may result from a machine learning algorithm identifying one or more underlying relationships within a dataset. In comparison to a physics-based model, such as Maxwell’s Equations, which are derived from first principles and define the mathematical relationship of a system, a pure machine learning model may not be derived from first principles. Once a machine learning model is developed, it may be queried in order to predict one or more outcomes for a given set of inputs. The type of input data used to query the model to create the prediction may correlate both in category and type to the dataset from which the model was developed.

[0074] The structure of, and the data contained within a dataset provided to a machine learning algorithm may vary depending on the intended function of the resulting machine learning model. The rows of data, or data points, within a dataset may contain one or more independent values. Additionally, datasets may contain corresponding dependent values. The independent values of a dataset may be referred to as “features,” and a collection of features may be referred to as a “feature space.” If dependent values are available in a dataset, they may be referred to as outcomes or “target values.” Although dependent values may be a component of a dataset for certain algorithms, not all algorithms utilize a dataset with dependent values. Furthermore, both the independent and dependent values of the dataset may comprise either numerical or categorical values.

[0075] While it may be true that machine learning model development is more successful with a larger dataset, it may also be the case that the whole dataset isn’t used to train the model. A test dataset may be a portion of the original dataset which is not presented to the algorithm for model training purposes. Instead, the test dataset may be used for what may be known as “model validation,” which may be a mathematical evaluation of how successfully a machine learning algorithm has learned and incorporated the underlying relationships within the original dataset into a machine learning model. This may include evaluating model performance according to whether the model is over-fit or under-fit. As it may be assumed that all datasets contain some level of error, it may be important to evaluate and optimize the model performance and associated model fit by a model validation. In general, the variability in model fit (e.g., whether a model is over-fit or under-fit) may be described by the “bias-variance trade-off.” As an example, a model with high bias may be an under-fit model, where the developed model is over-simplified, and has either not fully learned the relationships within the dataset or has over-generalized the

underlying relationships. A model with high variance may be an over-fit model which has overlearned about non-generalizable relationships within training dataset which may not be present in the test dataset. In a non-limiting example, these non-generalizable relationships may be driven by factors such as intrinsic error, data heterogeneity, and the presence of outliers within the dataset. The selected ratio of training data to test data may vary based on multiple factors, including, in a non-limiting example, the homogeneity of the dataset, the size of the dataset, the type of algorithm used, and the objective of the model. The ratio of training data to test data may also be determined by the validation method used, wherein some non-limiting examples of validation methods include k-fold cross-validation, stratified k-fold cross-validation, bootstrapping, leave-one-out cross-validation, resubstituting, random subsampling, and percentage hold-out.

[0076] In addition to the parameters that exist within the dataset, such as the independent and dependent variables, machine learning algorithms may also utilize parameters referred to as “hyperparameters.” Each algorithm may have an intrinsic set of hyperparameters which guide what and how an algorithm learns about the training dataset by providing limitations or operational boundaries to the underlying mathematical workflows on which the algorithm functions. Furthermore, hyperparameters may be classified as either model hyperparameters or algorithm parameters.

[0077] Model hyperparameters may guide the level of nuance with which an algorithm learns about a training dataset, and as such model hyperparameters may also impact the performance or accuracy of the model that is ultimately generated. Modifying or tuning the model hyperparameters of an algorithm may result in the generation of substantially different models for a given training dataset. In some cases, the model hyperparameters selected for the algorithm may result in the development of an over-fit or under-fit model. As such, the level to which an algorithm may learn the underlying relationships within a dataset, including the intrinsic error, may be controlled to an extent by tuning the model hyperparameters.

[0078] Model hyperparameter selection may be optimized by identifying a set of hyperparameters which minimize a predefined loss function. An example of a loss function for a supervised regression algorithm may include the model error, wherein the optimal set of hyperparameters correlates to a model which produces the lowest difference between the predictions developed by the produced model and the dependent values in the dataset. In addition to model hyperparameters, algorithm hyperparameters may also control the learning process of an algorithm, however algorithm hyperparameters may not influence the model performance. Algorithm hyperparameters may be used to control the speed and quality of the machine learning process. As such, algorithm hyperparameters may affect the computational intensity associated with developing a model from a specific dataset.

[0079] Machine learning algorithms, which may be capable of capturing the underlying relationships within a dataset, may be broken into different categories. One such category may include whether the machine learning algorithm functions using supervised, unsupervised, semi-supervised, or reinforcement learning. The objective of a supervised learning algorithm may be to determine one or more dependent variables based on their relationship to one or more independent variables. Supervised learning algorithms

are named as such because the dataset includes both independent and corresponding dependent values where the dependent value may be thought of as “the answer,” that the model is seeking to predict from the underlying relationships in the dataset. As such, the objective of a model developed from a supervised learning algorithm may be to predict the outcome of one or more scenarios which do not yet have a known outcome. Supervised learning algorithms may be further divided according to their function as classification and regression algorithms. When the dependent variable is a label or a categorical value, the algorithm may be referred to as a classification algorithm. When the dependent variable is a continuous numerical value, the algorithm may be a regression algorithm. In a non-limiting example, algorithms utilized for supervised learning may include Neural Networks, K-Nearest Neighbors, Naïve Bayes, Decision Trees, Classification Trees, Regression Trees, Random Forests, Linear Regression, Support Vector Machines (SVM), Gradient Boosting Regression, and Perception Back-Propagation.

[0080] The objective of unsupervised machine learning may be to identify similarities and/or differences between the data points within the dataset which may allow the dataset to be divided into groups or clusters without the benefit of knowing which group or cluster the data may belong to. Datasets utilized in unsupervised learning may not include a dependent variable as the intended function of this type of algorithm is to identify one or more groupings or clusters within a dataset. In a non-limiting example, algorithms which may be utilized for unsupervised machine learning may include K-means clustering, K-means classification, Fuzzy C-Means, Gaussian Mixture, Hidden Markov Model, Neural Networks, and Hierarchical algorithms.

[0081] In examples to determine a relationship using machine learning, a neural network (NN) 900, as illustrated in FIG. 9, may be utilized to model a three-dimensional finite element BHA to analyze lateral deflection experienced by BHA 116 (e.g., referring to FIG. 1) in both its lateral deflection in both inclination and pseudo-azimuth planes in a curved borehole 116 (e.g., referring to FIG. 1). FIG. 9 illustrates neural network (NN) 900. NN 900 may operate utilizing one or more information handling systems 120 (e.g., referring to FIG. 1) on computing network 800. Although a NN is illustrated, multiple models may be used with input output structures. These models may include flexible empirical models such as NN, gaussian processing methods, kriging methods, evolutionary methods such as genetic algorithms, classification methods, clustering methods empirical methods, or physics based methods such as equations of state, thermodynamic models, geological, geochemistry, or chemistry models, or kinetic models or any combinations therein including recursive combinations of similar or dissimilar models and iterative model combinations. A NN 900 is an artificial neural network with one or more hidden layers 902 between input layer 904 and output layer 906. In examples, NN 900 may be software on a single information handling system 120. In other examples, NN 900 may software running on multiple information handling systems 120 connected wirelessly and/or by a hard-wired connection in a network of multiple information handling systems 120. Herein, NN 900 may be applied in a wide array of implementations.

[0082] During operations, inputs **908** data are given to neurons **912** in input layer **904**. Neurons **912**, **914**, and **916** are defined as individual or multiple information handling

megaelectron volt (MeV). Energy resolution is in % at 662 kiloelectron volt (keV). The decay time is in nanoseconds (ns).

TABLE 1

Properties of Scintillators						
Material	Peak Emission (nm)	Light Yield (ph/MeV)	Density (g/cm ³)	Attenuation @ 1.5 MeV (cm ² /g)	Energy Resolution @ 662 keV	Decay Time (ns)
NaI(Tl)	415	38,000-55,000	3.67	114.7	7%	250
CsI(Tl)	540-550	52,000-65,000	4.51	140.9	6.90%	1,000
BGO	480	8,000-10,000	7.13	222.8	9.7-16%	300
LYSO(Ce)	420	30,000-33,000	7.1-7.2	225	8-20%	45
LaBr ₃	380	63,000	5.08-5.22	158.8	2.6-3.5%	16
CeBr ₃	380-390	57,000-66,000	5.1-5.2	159.4	3.8-4%	18-20

systems **120** connected in a computing network **800**. The output from neurons **912** may be transferred to one or more neurons **914** within one or more hidden layers **902**. Hidden layers **902** includes one or more neurons **914** connected in a network that further process information from neurons **912**. The number of hidden layers **902** and neurons **912** in hidden layer **902** may be determined by personnel that designs NN **900**. Hidden layers **902** is defined as a set of information handling system **120** assigned to specific processing. Hidden layers **902** spread computation to multiple neurons **912**, which may allow for faster computing, processing, training, and learning by NN **900**. Output from NN **900** may be computed by neurons **916**. An information handling system **120** (e.g., referring to FIG. 1) being utilized in a computing network **800**, NN **900**, or alone may control measurement operations downhole with pulsed neutron logging tool **132**. Specifically, information regarding a neutron to gamma ray timing may be computed and utilized by information handling system **120** during measurement operations in which pulsed neutron generator **308** (e.g., referring to FIG. 3) is pulsed. In a sub-us time domain, fast neutrons and inelastic gamma rays **320** dominate, whereas in a 10-1000 us time range, thermal neutrons **328** are bouncing around and capture gamma rays are detected.

[0083] As mentioned, embodiments herein of a pulsed neutron logging tool include the material CeBr₃ as the scintillator. Table 1 below compares CeBr₃ with commonly used scintillators in the well logging industry. As indicated in Table 1, CeBr₃ provides for high density, high light yield, fast decay time, good energy resolution, and elevated temperature stability, and little to no intrinsic radiation background (see FIG. 10). Such properties can generally make CeBr₃ a good scintillator for nuclear logging products including new products and as an upgrade to existing products in implementations.

[0084] Table 1 lists CeBr₃ along with common scintillators. The common scintillators in Table 1 are sodium iodide doped with thallium [NaI(Tl)], cesium iodide doped with thallium [CsI(Tl)], bismuth germanium oxide (BGO) (Bi₄Ge₃O₁₂) (also known as bismuth germanate), lutetium yttrium oxyorthosilicate doped with cerium [LYSO (Ce)], and lanthanum bromide (LaBr₃). The peak emission is in nanometers (nm). The light yield is in photons per megaelectron volt (ph/MeV). The density is in grams per cubic centimeter (g/cm³). Attenuation is the mass attenuation coefficient in square centimeters per gram (cm²/g) at 1.5

[0085] FIG. 10 is a plot **1000** of intrinsic background spectra (outputs from the photomultiplier tube of a scintillator captured with a pulse height and multichannel analyzer) in counts/second (sec)/cm²/keV versus energy in keV. The intrinsic background spectra may also be labeled as intrinsic activity spectra.

[0086] The curve **1002** is an intrinsic background spectrum for NaI(Tl). The curve **1004** is an intrinsic background spectrum for LaBr₃ doped with 5% Ce (LaBr₃:5% Ce). The curve **1006** is an intrinsic background spectrum for CeBr₃.

[0087] FIG. 11 is plot **1100** of natural gamma-ray spectra (K-U-Th) of NaI and CeBr₃ in normalized counts versus energy in MeV. Six curves (spectra) are plotted. For NaI, three curves (spectra) are given for naturally occurring radioactive elements potassium (K), uranium (U), and thorium (Th), respectively, that give off gamma rays. Likewise, for CeBr₃, three curves (spectra) are given for K, U, and Th, respectively. A legend **1102** for the six curves (spectra) is depicted.

[0088] In implementations, CeBr₃ can replace NaI scintillators in existing logging products. In implementations, CeBr₃ as a scintillator can have advantages of relatively high resolution and low background noise to provide dual spectroscopy measurements (neutron-induced gamma ray spectroscopy and natural gamma-ray spectroscopy) in a single tool (system). The higher resolution may help with accurate spectral deconvolution in both applications (neutron-induced gamma ray spectroscopy and natural gamma-ray spectroscopy). Further, the CeBr₃ low-to-no intrinsic radiation background may facilitate the natural gamma-ray measurement. Such dual spectroscopy cannot generally be achieved, for example, by a higher resolution conventional detector like LaBr₃ because the LaBr₃ high intrinsic background lowers the signal-to-noise ratio of the system and makes the measurement too imprecise.

[0089] FIG. 12 is a plot **1200** of a neutron induced capture gamma-ray spectrum from a logging while drilling (LWD) tool equipped with CeBr₃ as a scintillator in a dolomite formation. The plot is in counts per second (CPS) per channel (of a multichannel analyzer for gamma ray spectroscopy) versus energy in MeV. The LWD tool for downhole is a pulsed neutron logging tool having a spectrum analyzer in which the spectroscopy analysis is performed downhole on board the pulsed neutron logging tool. In other implementations, pulsed neutron logging tool can transmit raw spectral data and the spectroscopy analysis performed remotely at surface and/or in the cloud. Applicable pulsed

neutron logging tool can be analogous to the pulsed neutron logging tools of FIGS. 3 and 13A-C, and other configurations of a pulsed neutron logging tool.

[0090] FIGS. 13A-C are downhole logging tools 1300A-C each as a pulsed neutron logging tool and each having a gamma ray scintillator detector 1302 with CeBr_3 as a scintillator. Downhole logging tools 1300A-B can be for LWD operations. Downhole logging tool 1300C can be for wireline operations, or for through-the-bit intervention/drilling operations, or with a suitable conveyance platform. The downhole logging tools 1300A-C have a gamma ray scintillator detector 1302 and a pulsed neutron generator (PNG) 1304, as does the pulsed neutron logging tool 132 of FIG. 3. The downhole logging tools 1300A-C each include the PNG 1304 that may be analogous to the PNG 308 (having neutron tube 312) of the pulsed neutron logging tool 132 of FIG. 3. As indicated in FIGS. 13A-C, the scintillator detector 1302 and the PNG 1304 may be within a body 1306 of the logging tool 1300A-C. The PNG 1304 and detector 1302 position can be swapped axially. The number of detectors 1302 in each tool 1300A-C can be more than one in the radial and/or axial direction.

[0091] The logging tool 1300A (FIG. 13A) and the logging tool 1300B (FIG. 13B) may each be a LWD tool that may be applied, for example, as a component of a bottom-hole assembly of a drillstring. While drilling into a subterranean formation, the PNG 1304 may be in operation of pulsed mode, and the tool may provide neutron induced gamma-ray spectroscopy derived measurements, such as elemental weight fraction for mineralogy of the subterranean formation. While tripping in, tripping out or in a desired interval (such as wipe run), the PNG 1304 may be turned OFF for a prolonged period and thus may provide natural gamma-ray spectroscopy related measurements, such as K, U, and Th concentrations, and so on. Long decay formation activation and mud activation may be accounted for. The data processing procedures or work-flow of the downhole processing, by a processor and programmed memory on the logging tool 1300A/B, may be switched for desired gamma-ray spectroscopy measurements, among inelastic and capture gamma rays, long decay activation and natural gamma rays. This can be achieved, for example, by downlink command or automatically determined by system operation flags, such as PNG 1304 voltage (and current), detector 1302 counting rate, and so forth. With a downlink command, a human or a processor, such as information handling system 120 running an artificial intelligence NN 900 software, decides to switch the downhole processing work-flow and sends that instruction to the tool 1300A/B from surface by a downlink command. For automated operation, a determination is performed by the downhole processor and programmed memory on the logging tool 1300A/B.

[0092] The logging tools 1300A and 1300B each have a flow path 1308 as depicted within the body 1306 for drilling mud (drilling fluid). The flow path 1308 of the logging tool 1300A can be radially offset to one side (as depicted) or radially centered. The flow path 1308 configuration of the logging tool 1300B can be characterized as a sonde/probe based mechanical layout.

[0093] The logging tool 1300C (FIG. 13C) does not have a flow path (mud bore) for drilling mud (drilling fluid) within the body 1306. The logging tool 1300C may be a wireline tool or other tools in which drilling fluid (mud) is not routed through the tool body, such as a through-the-bit

intervention operations. In those implementations, instead of drilling and tripping in/out, the downhole logging tool 1300 (e.g., as a wireline tool applied into a borehole via a wireline) can, for example, log one kind of spectroscopy measurements when tripping down the borehole and log the other spectroscopy measurements when tripping up the borehole, saving rig time. The two different kinds of spectroscopy measurements logged may be, for example, for (1) neutron-induced gamma rays and (2) natural gamma rays. A through-the-bit application or through-the-bit methodology is referred as a conveyance platform for the logging tool 1300C to access lateral measurements in some of difficult-to-access wells.

[0094] The techniques may include comprising switching (alternating) between two or modes (operating modes, measuring modes, generating modes, spectral modes) of the pulsed neutron logging tool. In some implementations, two modes include a pulsed neutron logging (PNL) mode and a natural spectral gamma ray (NSGR) logging mode.

[0095] Logging while drilling (LWD), measurement while drilling (MWD), pressure while drilling (PWD), and the like, may refer to techniques for gathering downhole data while drilling without removing the drillstring from the wellbore.

[0096] In an LWD or MWD implementation of the logging tools 1300A and 1300B, the mode (operating mode, measuring mode) of the tool 1300A and tool 1300B may switch between [1] a PNL mode and [2] a NSGR mode. The switch between these two modes may be achieved, for example, by sending downlink commands from the surface using wired drilling pipe (wired drillstring). Mud pulse telemetry may be transmitting LWD or MWD data acquired downhole to surface, using pressure pulses in the mud (drilling fluid) system. Wired drilling pipe (wired drill pipe) may provide for bi-directional communication between a computer at surface and the tools 1300A-B downhole. The switch between the PNL mode and the NSGR mode may be achieved, for example, by sending downlink commands from the surface using pipe movement (rotation or up/down combinations) to encode switch signals. For instance, a downhole processor within the logging tool 1300 A/B equipped with a motion sensor may decode the switch signals to determine a switch between the two modes. The switch between the two modes may be achieved, for example, by downhole algorithm that determines automatically when to be in PNL mode or NSGR mode. The algorithm may include or consider, for example, system operation flags, such as PNG 1304 voltage (and current), neutron yield, detector 1302 total counting rate, or a particular gamma-ray counting rate within a certain energy window in the spectrum, or a particular gamma-ray counting rates in a certain timing window in a given pulsing cycle, and so forth. The algorithm may be a code (logic, instructions) stored in memory of the tools 1300A-B and performed (executed, implemented) by a hardware processor on the tools 1300A-B.

[0097] In the PNL mode in the LWD or MWD implementations, the generator ON/OFF timing scheme may be optimized or adjusted beneficially to include a repetitive background measuring period, e.g., an idle or prolonged OFF state within the pulsing cycle. The data collected during each background measuring period may be summed periodically to form natural spectral gamma-ray measurements. Thus, the

PNL mode and NSGR mode can be carried out simultaneously or contemporaneously in one timing scheme during a single logging trip.

[0098] In one example of an LWD or MWD implementation, such as with tool **1300A** of FIG. **13A**, the position of detector **1302** with CeBr_3 is offset in the tool from the radial center so that the detector can be azimuthally sensitive. When the tool **1300A** rotates during drilling, the detector **1302** senses different sections of the formation **204** and acquires different natural gamma rays from different sections of formation **204** azimuthally in the borehole. A natural gamma image may be produced in the NSGR mode to facilitate geosteering. Geosteering may be the beneficial or optimal placement of a wellbore based on the results of real time downhole geological and geophysical logging measurements rather than three-dimensional targets in space. An objective may be, for example, to keep a directional wellbore within a hydrocarbon pay zone defined in terms of its resistivity, density or even biostratigraphy. In drilling a borehole, geosteering may be the act of adjusting the borehole position (e.g., inclination and azimuth angles) on the fly to reach one or more geological targets. These changes may be based on geological information gathered while drilling.

[0099] In examples, the mode switching sequence can be set to as NSGR mode to geosteering each drilling stand as they drill, and then downlink the command to enable PNL mode and wipe back up (e.g., slowly) to log pulsed neutron spectroscopy or visa versa, and repeat this process over the zones of interest. An advantage may be to geosteer with NSGR most of the time (a majority of time the tool is operating downhole) and PNL when the tool is in a zone where pulse neutron spectroscopy is beneficial or necessary.

[0100] Moreover, implementations of tool **1300C** include through-bit applications in which the tool is similar to a wireline tool but carried downhole by the drillstring and not deployed with a wireline. Once drilling is completed, the wireline tool may be released through the drilling bit and log data while tripping out.

[0101] An embodiment is a pulsed neutron logging tool (e.g., as a LWD tool, wireline tool, etc.) for a borehole in a subterranean formation, including a pulsed neutron generator to broadcast neutrons into the subterranean formation, and a gamma ray scintillator detector having a scintillator including CeBr_3 . The pulsed neutron generator may include a neutron tube to generate the neutrons for broadcasting. The scintillator may be a scintillation crystal including the CeBr_3 . The gamma ray scintillator detector may include a photomultiplier tube coupled to the scintillator. The pulsed neutron logging tool is configured to detect (measure) neutron-induced gamma rays from the subterranean formation and natural gamma rays from the subterranean formation. In implementations, shielding is disposed between the gamma ray scintillator detector and the pulsed neutron generator. The pulsed neutron logging tool may have an outer housing (e.g., metal) in which the pulsed neutron generator and the gamma ray scintillator detector are disposed in the outer housing. In implementations, the pulsed neutron logging tool includes an inner housing (e.g., metal) disposed in the outer housing for an insulating gas (e.g., SF_6), wherein the pulsed neutron generator is disposed within the inner housing. For implementations of the pulsed neutron logging tool as a LWD tool, the LWD tool may have a body having a flow path for drilling fluid.

[0102] Another embodiment is a method of deploying a pulsed neutron logging tool, including lowering the pulsed neutron logging tool into a borehole in a subterranean formation, and detecting natural gamma rays from the subterranean formation via a scintillator detector of the pulsed neutron logging tool. The scintillator detector includes a scintillator including CeBr_3 . In implementations, the gamma ray scintillator detector includes a photomultiplier tube coupled to the scintillator. The method includes broadcasting neutrons into the subterranean formation via a pulsed neutron generator of the pulsed neutron logging tool. In implementations, the method includes generating the neutrons via a neutron tube of the pulsed neutron generator. The method includes detecting neutron-induced gamma rays from the subterranean formation via the scintillator detector. In implementations, the pulsed neutron generator and the gamma ray scintillator detector are disposed in an outer housing of the pulsed neutron logging tool. In implementations, a shielding is disposed between the gamma ray scintillator detector and the pulsed neutron generator. In implementations, the pulsed neutron logging tool is lowered into the borehole via a wireline. In other implementations, the pulsed neutron logging tool is an LWD tool, wherein lowering the pulsed neutron logging tool includes lowering the pulsed neutron logging tool as a component of a drillstring into the borehole. In particular, the pulsed neutron logging tool may be a component of a bottom-hole assembly of the drillstring. Moreover, the pulsed neutron logging tool as a LWD tool may have a body having a flow path for drilling fluid, and the method may include drilling the subterranean formation via the drilling fluid and a drill bit at an end of the drillstring to form the borehole.

[0103] The method may include performing in the borehole, via the pulsed neutron logging tool, spectroscopy measurements of natural gamma rays, pulsed neutron induced inelastic gamma rays, and pulsed neutron induced capture gamma rays. The method may include switching between a mode (a first mode) of the pulsed neutron logging tool in which spectroscopy measurements of natural gamma rays are performed and another mode (a second mode) of the pulsed neutron logging tool in which spectroscopy measurement of neutron induced inelastic gamma rays and neutron induced capture gamma rays are performed. The switching may be, for example, (directed) by downlink command or by automatically determining the switching by system operation flags of the pulsed neutron logging tool. The system operation flags may include, for example, voltage of the pulsed neutron generator, current of the pulsed neutron generator, neutron yield or counting rate of the one or more scintillator detectors, counting rate of an energy window of the one of more scintillator detectors, or counting rate of a timing window of the one of more scintillator detectors during a pulsing sequence, or any combinations thereof. The method may include switching between two or more modes of the pulsed neutron logging tool. The two modes may include, for example, a PNL mode and a NSGR logging mode. The method may include implementing repetitive background measuring periods (repetitive/prolonged background measuring periods) in the PNL mode for an ON/OFF timing scheme of the pulsed neutron generator; and summing data collected during each background measuring period to generate a natural spectral gamma-ray response for

an azimuthal analysis. The method may include generating a natural gamma image by the pulsed neutron logging tool to facilitate geosteering.

[0104] The methods and systems described above are an improvement over the conventional technology as the methods and systems described herein provide dual purpose gamma-ray spectroscopy via a downhole logging tool having a scintillator that is or includes cerium bromide (CeBr_3). The systems and methods may comprise any of the various features disclosed herein, comprising one or more of the following statements.

[0105] Statement 1. A pulsed neutron logging tool for a borehole in a subterranean formation, comprising: a pulsed neutron generator to broadcast neutrons into the subterranean formation; and a gamma ray scintillator detector comprising cerium bromide (CeBr_3), wherein the pulsed neutron logging tool is configured to detect neutron-induced gamma rays from the subterranean formation and natural gamma rays from the subterranean formation.

[0106] Statement 2. The pulsed neutron logging tool of Statement 1, wherein the pulsed neutron logging tool comprises more than one gamma ray scintillator detector comprising CeBr_3 , and wherein each gamma ray scintillator detector comprises a scintillation crystal comprising the CeBr_3 .

[0107] Statement 3. The downhole tool of any preceding Statement, wherein the pulsed neutron logging tool is a logging while drilling (LWD) tool, and wherein the LWD tool comprises a body having a flow path for drilling fluid.

[0108] Statement 4. The pulsed neutron logging tool of any preceding Statement, wherein the pulsed neutron logging tool is a wireline tool, or wherein the pulsed neutron logging tool is a tool not configured to route drilling fluid through a tool body of the tool, or a combination thereof.

[0109] Statement 5. The pulsed neutron logging tool of any preceding Statement, wherein the scintillator detector comprises: a scintillator comprising the CeBr_3 ; and a photomultiplier tube coupled to the scintillator.

[0110] Statement 6. The pulsed neutron logging tool of any preceding Statement, wherein shielding (a shield) is disposed between the gamma ray scintillator detector and the pulsed neutron generator, and wherein the pulsed neutron generator comprises a neutron tube to generate the neutrons for broadcasting.

[0111] Statement 7. The pulsed neutron logging tool of any preceding Statement, comprising an outer housing comprising metal, wherein the pulsed neutron generator and the gamma ray scintillator detector are disposed in the outer housing.

[0112] Statement 8. The pulsed neutron logging tool of Statement 7, comprising an inner housing for an insulating gas, the inner housing comprising metal and disposed in the outer housing, wherein the pulsed neutron generator is disposed within the inner housing.

[0113] Statement 9. The pulsed neutron logging tool of Statement 8, wherein the insulating gas comprises silicon hexafluoride (SF_6).

[0114] Statement 10. A method of deploying a pulsed neutron logging tool, comprising: lowering the pulsed neutron logging tool into a borehole in a subterranean formation; detecting natural gamma rays from the

subterranean formation via one or more scintillator detectors of the pulsed neutron logging tool, wherein the one or more scintillator detectors comprise cerium bromide (CeBr_3); broadcasting neutrons into the subterranean formation via a pulsed neutron generator of the pulsed neutron logging tool; and detecting neutron-induced gamma rays and natural gamma rays from the subterranean formation via the one or more scintillator detectors.

[0115] Statement 11. The method of Statement 10, comprising drilling the subterranean formation via a drill bit at an end of a drillstring to form the borehole, wherein the pulsed neutron logging tool is a LWD tool, and wherein lowering the pulsed neutron logging tool into the borehole comprises lowering the pulsed neutron logging tool as a component of the drillstring into the borehole.

[0116] Statement 12. The method of Statement 11, wherein the LWD tool comprises a body having a flow path for drilling fluid.

[0117] Statement 13. The method of Statement 10, wherein lowering the pulsed neutron logging tool comprises lowering the pulsed neutron logging tool via a wireline or through-the-bit methodology into the borehole.

[0118] Statement 14. The method of Statements 10-13, wherein the one or more gamma ray scintillator detectors each comprise: a scintillator comprising the CeBr_3 ; and a photomultiplier tube coupled to the scintillator.

[0119] Statement 15. The method of Statements 10-14, comprising generating the neutrons via a neutron tube of the pulsed neutron generator, wherein the pulsed neutron generator and the one or more gamma ray scintillator detectors are disposed in an outer housing of the pulsed neutron logging tool, and wherein a shielding is disposed between the one or more gamma ray scintillator detectors and the pulsed neutron generator.

[0120] Statement 16. The method of Statements 10-15, comprising performing in the borehole, via the pulsed neutron logging tool, spectroscopy measurements of natural gamma rays, pulsed neutron induced inelastic gamma rays, and pulsed neutron induced capture gamma rays.

[0121] Statement 17. The method of Statements 10-16, comprising switching between a mode of the pulsed neutron logging tool in which spectroscopy measurements of natural gamma rays are performed and another mode of the pulse neutron logging tool in which spectroscopy measurement of neutron induced inelastic gamma rays and neutron induced capture gamma rays are performed, the switching by downlink command or by automatically determining the switching by system operation flags of the pulsed neutron logging tool.

[0122] Statement 18. The method of Statements 10-17, comprising switching between two modes of the pulsed neutron logging tool, the two modes comprising a PNL mode and a NSGR logging mode.

[0123] Statement 19. The method of Statement 18, comprising: implementing repetitive background measuring periods in the PNL mode for an ON/OFF timing scheme of the pulsed neutron generator; and summing

data collected during the background measuring periods to generate a natural spectral gamma-ray response for an azimuthal analysis.

[0124] Statement 20. The method of Statements 10-19, comprising generating a natural gamma image by the pulsed neutron logging tool to facilitate geosteering.

[0125] As it is impracticable to disclose every conceivable embodiment of the technology described herein, the figures, examples, and description provided herein disclose only a limited number of potential embodiments. One of ordinary skill in the art would appreciate that any number of potential variations or modifications may be made to the explicitly disclosed embodiments, and that such alternative embodiments remain within the scope of the broader technology. Accordingly, the scope should be limited only by the attached claims. Further, the compositions and methods are described in terms of “comprising,” “containing,” or “including” various components or steps, the compositions and methods may also “consist essentially of” or “consist of” the various components and steps. Moreover, the indefinite articles “a” or “an,” as used in the claims, are defined herein to mean one or more than one of the elements that it introduces. Certain technical details, known to those of ordinary skill in the art, may be omitted for brevity and to avoid cluttering the description of the novel aspects.

[0126] For further brevity, descriptions of similarly named components may be omitted if a description of that similarly named component exists elsewhere in the application. Accordingly, any component described with respect to a specific figure may be equivalent to one or more similarly named components shown or described in any other figure, and each component incorporates the description of every similarly named component provided in the application (unless explicitly noted otherwise). A description of any component is to be interpreted as an optional embodiment which may be implemented in addition to, in conjunction with, or in place of an embodiment of a similarly-named component described for any other figure.

[0127] As used herein, adjective ordinal numbers (e.g., first, second, third, etc.) are used to distinguish between elements and do not create any particular ordering of the elements. As an example, a “first element” is distinct from a “second element”, but the “first element” may come after (or before) the “second element” in an ordering of elements. Accordingly, an order of elements exists only if ordered terminology is expressly provided (e.g., “before”, “between”, “after”, etc.) or a type of “order” is expressly provided (e.g., “chronological”, “alphabetical”, “by size”, etc.). Further, use of ordinal numbers does not preclude the existence of other elements. As an example, a “table with a first leg and a second leg” is any table with two or more legs (e.g., two legs, five legs, thirteen legs, etc.). A maximum quantity of elements exists only if express language is used to limit the upper bound (e.g., “two or fewer”, “exactly five”, “nine to twenty”, etc.). Similarly, singular use of an ordinal number does not imply the existence of another element. As an example, a “first threshold” may be the only threshold and therefore does not necessitate the existence of a “second threshold”.

[0128] As used herein, the word “data” may be used as an “uncountable” singular noun-not as the plural form of the singular noun “datum”. Accordingly, throughout the application, “data” is generally paired with a singular verb (e.g., “the data is modified”). However, “data” is not redefined to

mean a single bit of digital information. Rather, as used herein, “data” means any one or more bit(s) of digital information that are grouped together (physically or logically). Further, “data” may be used as a plural noun if context provides the existence of multiple “data” (e.g., “the two data are combined”).

[0129] As used herein, the term “operative connection” (or “operatively connected”) means the direct or indirect connection between devices that allows for interaction in some way (e.g., via the exchange of information). For example, the phrase ‘operatively connected’ may refer to a direct connection (e.g., a direct wired or wireless connection between devices) or an indirect connection (e.g., multiple wired and/or wireless connections between any number of other devices connecting the operatively connected devices).

What is claimed is:

1. A pulsed neutron logging tool for a borehole in a subterranean formation, comprising:

- a pulsed neutron generator to broadcast neutrons into the subterranean formation; and
- a gamma ray scintillator detector comprising cerium bromide (CeBr_3), wherein the pulsed neutron logging tool is configured to detect neutron-induced gamma rays from the subterranean formation and natural gamma rays from the subterranean formation.

2. The pulsed neutron logging tool of claim 1, wherein the pulsed neutron logging tool comprises more than one gamma ray scintillator detector comprising CeBr_3 , and wherein each gamma ray scintillator detector comprises a scintillation crystal comprising the CeBr_3 .

3. The pulsed neutron logging tool of claim 1, wherein the pulsed neutron logging tool is a logging while drilling (LWD) tool, and wherein the LWD tool comprises a body having a flow path for drilling fluid.

4. The pulsed neutron logging tool of claim 1, wherein the pulsed neutron logging tool is a wireline tool, or wherein the pulsed neutron logging tool is a tool not configured to route drilling fluid through a tool body of the tool, or a combination thereof.

5. The pulsed neutron logging tool of claim 1, wherein the gamma ray scintillator detector comprises:

- a scintillator comprising the CeBr_3 ; and
- a photomultiplier tube coupled to the scintillator.

6. The pulsed neutron logging tool of claim 1, wherein shielding is disposed between the gamma ray scintillator detector and the pulsed neutron generator, and wherein the pulsed neutron generator comprises a neutron tube to generate the neutrons for broadcasting.

7. The pulsed neutron logging tool of claim 1, comprising an outer housing comprising metal, wherein the pulsed neutron generator and the gamma ray scintillator detector are disposed in the outer housing.

8. The pulsed neutron logging tool of claim 7, comprising an inner housing for an insulating gas, the inner housing comprising metal and disposed in the outer housing, wherein the pulsed neutron generator is disposed within the inner housing.

9. The pulsed neutron logging tool of claim 8, wherein the insulating gas comprises silicon hexafluoride (SF_6).

10. A method of deploying a pulsed neutron logging tool, comprising:

- lowering the pulsed neutron logging tool into a borehole in a subterranean formation;

detecting natural gamma rays from the subterranean formation via one or more scintillator detectors of the pulsed neutron logging tool, wherein the one or more scintillator detectors comprise cerium bromide (CeBr_3);

broadcasting neutrons into the subterranean formation via a pulsed neutron generator of the pulsed neutron logging tool; and

detecting neutron-induced gamma rays and natural gamma rays from the subterranean formation via the one or more scintillator detectors.

11. The method of claim **10**, comprising drilling the subterranean formation via a drill bit at an end of a drillstring to form the borehole, wherein the pulsed neutron logging tool is a logging while drilling (LWD) tool, and wherein lowering the pulsed neutron logging tool into the borehole comprises lowering the pulsed neutron logging tool as a component of the drillstring into the borehole.

12. The method of claim **11**, wherein the LWD tool comprises a body having a flow path for drilling fluid.

13. The method of claim **10**, wherein lowering the pulsed neutron logging tool comprises lowering the pulsed neutron logging tool via a wireline or through-the-bit methodology into the borehole.

14. The method of claim **10**, wherein the one or more scintillator detectors each comprise:

a scintillator comprising the CeBr_3 ; and

a photomultiplier tube coupled to the scintillator.

15. The method of claim **10**, comprising generating the neutrons via a neutron tube of the pulsed neutron generator, wherein the pulsed neutron generator and the one or more scintillator detectors are disposed in an outer housing of the

pulsed neutron logging tool, and wherein a shielding is disposed between the one or more scintillator detectors and the pulsed neutron generator.

16. The method of claim **10**, comprising performing in the borehole, via the pulsed neutron logging tool, spectroscopy measurements of natural gamma rays, pulsed neutron induced inelastic gamma rays, and pulsed neutron induced capture gamma rays.

17. The method of claim **10**, comprising switching between a mode of the pulsed neutron logging tool in which spectroscopy measurements of natural gamma rays are performed and another mode of the pulsed neutron logging tool in which spectroscopy measurement of neutron induced inelastic gamma rays and neutron induced capture gamma rays are performed, the switching by downlink command or by automatically determining the switching by system operation flags of the pulsed neutron logging tool.

18. The method of claim **10**, comprising switching between two modes of the pulsed neutron logging tool, the two modes comprising a pulsed neutron logging (PNL) mode and a natural spectral gamma ray (NSGR) logging mode.

19. The method of claim **18**, comprising:

implementing repetitive background measuring periods in the PNL mode for an ON/OFF timing scheme of the pulsed neutron generator; and

summing data collected during the repetitive background measuring periods to generate a natural spectral gamma-ray response for an azimuthal analysis.

20. The method of claim **18**, comprising generating a natural gamma image by the pulsed neutron logging tool to facilitate geosteering.

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